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Joint Price–Volume Anomalies in an Indonesian Mining Equity: Statistical Association and Contextual Evidence from BRMS.JK, 2016–2025

Research Article

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Abstract

This study examines extreme price and trading-volume behavior in PT Bumi Resources Minerals Tbk (BRMS) during 2016–2025 using an integrated descriptive-associational and contextual framework. Daily observations from Yahoo Finance were screened and chronologically aligned, yielding 2,390 retained trading observations and 2,389 valid paired observations. Price anomalies were identified from the upper empirical tail of absolute daily returns, while volume anomalies were classified from the lower and upper tails of signed daily volume changes. Twenty-three simultaneous price–volume anomalies were observed, compared with approximately 12.06 expected under statistical independence; 20 combined a sharp price increase with an abnormally large volume increase. Pearson's chi-square test indicated significant dependence ($\chi^2 = 11.6304$, $p = 0.0006$), and the sample odds ratio was 2.24 (95% CI: 1.39–3.61). Phi/Cramér's V was approximately 0.070, indicating limited association magnitude. Documentary validation of 11 prespecified events classified 2 as High, 4 as Moderate, and 5 as Not Externally Validated. The findings distinguish aggregate statistical dependence from event-specific documentary correspondence. Joint price–volume anomalies may support transparent market screening and structured follow-up analysis, but they should not be interpreted as standalone evidence of prediction, causality, or market misconduct in practice.

Keywords: price–volume association; extreme returns; trading volume; market anomaly; mining equity; Indonesia

1. Introduction

Stock prices and trading volume provide complementary views of market activity: prices summarize valuation changes, whereas volume records the intensity of trading. Their joint behavior has therefore been central to empirical finance. Karpoff (1987), Gallant et al. (1992), and Campbell et al. (1993) document systematic price-volume relationships, but those relationships do not imply a single or stable causal mechanism. For the present study, joint analysis is used to identify empirical dependence and unusual co-occurrences while maintaining a strict distinction between statistical association and causal attribution.

A tail-focused approach is particularly useful when average relations may conceal infrequent but economically visible observations. The present design therefore treats the price and volume dimensions in parallel at the classification stage while recognizing that their empirical distributions require different directional rules. Price extremes are identified from absolute return magnitude and then restored to their signed direction, whereas volume extremes are identified from the signed distribution because sharp contractions and expansions are substantively different trading states. This design choice allows the analysis to preserve direction without assuming that positive and negative tails have identical economic meaning.

Return-volume dependence can vary across market states and distributional regions. Ciner (2015) reports time-varying return-volume relationships in precious-metals mining stocks, while Lin and González-Rivera

(2019) show that extreme returns and trading intensity can display dynamics that differ from those near the center of the distribution. This heterogeneity motivates a tail-oriented design in which unusually large price and volume observations are identified explicitly rather than inferred from unconditional averages.

Extreme observations are a recognized object of market research. Gervais et al. (2001) examine unusual trading-volume states, and Larson and Madura (2003) distinguish extreme one-day returns associated with identifiable public information from events lacking a readily documented catalyst. Because the literature uses different definitions of 'extreme', the present study does not treat any percentile as a universal market standard. Instead, empirical percentiles provide transparent, sample-specific operational boundaries for identifying unusual BRMS observations.

Quantitative co-occurrence alone does not reveal the public-information environment surrounding an event. Research on attention and information dissemination shows that news, media exposure, extreme returns, and unusual volume can be related to trading behavior (Barber & Odean, 2008; Engelberg & Parsons, 2011; Peress, 2014). Those studies motivate a separate documentary layer here, but their causal results are not transferred to this security-level design. Public information is used only to assess temporal and substantive correspondence with selected anomalies.

Mining equities offer a relevant setting because equity-market, operational, financing, and commodity-related conditions may interact without making the stock equivalent to the underlying commodity. Baur et al. (2021), Ciner (2015), and Qin et al. (2023) document distinctive risk and tail behavior in mining-related equities. BRMS is therefore examined as a resource-sector security with security-specific price-volume dynamics; the study does not estimate structural commodity exposure or use BRMS as a proxy for gold-price effects.

The Indonesian mining-equity literature reviewed for this study includes macro-financial, commodity, forecasting, portfolio, and risk applications (Ahadiat et al., 2023; Endri et al., 2021; Pungkasanti et al., 2026; Putra & Robiyanto, 2019). Comparatively less evidence was located on a framework that combines long-horizon price-volume characterization, empirical tail classification, contemporaneous joint-anomaly mapping, categorical association testing, and structured documentary validation. This observation is limited to the reviewed literature and is not a claim that Indonesian mining research is methodologically homogeneous.

The emerging-market setting also matters for interpretation. Evidence from one Indonesian issuer cannot establish a market-wide regularity, but it can reveal how established price-volume concepts behave in a security with distinctive trading history and episodic changes in activity. By focusing on a long daily window rather than a short event interval, the study first characterizes the underlying distribution of BRMS observations and then identifies unusual co-occurrences relative to that same empirical history. The contextual stage is deliberately secondary to the statistical classification, which reduces the risk that event narratives determine which observations are labeled anomalous.

At the issuer level, BRMS has been examined in other academic contexts, so it should not be described as an unstudied company. Under the search strategy used for this study, however, no directly comparable peer-reviewed BRMS paper was identified that integrates the full sequence of tail identification, same-observation price-volume mapping, association magnitude, and prespecified contextual validation. The contribution is therefore analytical integration and BRMS-specific emerging-market evidence rather than methodological invention or a 'first-ever' claim.

This contribution is therefore narrower than proposing a new anomaly test and broader than presenting a simple descriptive chronology. The paper links four analytical questions that are often separated: what constitutes an extreme observation within the sample; whether price and volume extremes occur on the same day; whether their binary statuses depart from independence; and whether selected joint events can be matched to credible public information under a fixed search protocol. The resulting evidence is intended to be reproducible and auditable while remaining explicit about the limits of a single-security observational design.

Accordingly, the study examines BRMS daily price and trading-volume dynamics from 2016 to 2025 with four connected objectives: to characterize long-horizon price and trading activity; identify empirical price and signed-volume tail events; evaluate their contemporaneous co-occurrence and statistical association; and contextualize a prespecified subset of joint anomalies using credible public information. The design deliberately separates descriptive dynamics, tail classification, statistical dependence, and documentary correspondence so that neither association nor contextual evidence is interpreted as causal transmission.

2. Materials and Methods

2.1. Research Design and Analytical Framework

This study employed a longitudinal, single-security, descriptive-associational design for PT Bumi Resources Minerals Tbk (BRMS.JK), 2016–2025. The workflow moved sequentially from data screening and chronological alignment to construction of price and volume variables, empirical tail classification, same-observation joint-anomaly mapping, 2×2 association testing, and documentary validation of a prespecified event subset. Joint price-volume analysis is consistent with established empirical finance (Campbell et al., 1993; Gallant et al., 1992; Karpoff, 1987), but the present design does not estimate causal effects, lead-lag transmission, or expected-return models.

2.2. Data Source, Security, and Observation Period

The empirical analysis focused on PT Bumi Resources Minerals Tbk (BRMS.JK). Daily observations were obtained from the Yahoo Finance historical-data interface (Yahoo Finance, n.d.), which is treated as a secondary data-access source rather than the official Indonesia Stock Exchange database. The source fields included Date, Open, High, Low, Close, Adjusted Close, and Volume; calculations used the Close field and therefore represent simple close-to-close price returns rather than an explicit total-return measure. The retained observation period was 4 January 2016–30 December 2025, yielding 2,390 trading observations and 2,389 valid lagged pairs. The archived analytical record shows processing on 5 August 2026 at 07:03:57; software package versions were not reconstructed retrospectively because they were not preserved in the metadata.

The source snapshot was preserved as an MHTML copy of the Yahoo Finance historical-data page, providing a fixed archival input for the analysis rather than relying on a subsequently changing web view. The analytical workflow separated the source table, retained clean observations, excluded-row audit, calculated variables, metadata, and methodological settings. Missing or invalid volume observations were preserved as missing in the audit and were never converted to zero, which is important because zero would represent a substantive trading value rather than an unavailable observation.

2.3. Data Preparation and Trading-Day Alignment

Observations were screened for valid dates, positive numerical market fields, duplicate or non-trading records, and logical OHLC consistency. Missing or invalid Volume values were never replaced with zero; such rows were retained in the audit trail but excluded before lagged price and volume variables were constructed. Retained observations were sorted from oldest to newest before applying a one-observation lag, so $t-1$ denotes the immediately preceding retained trading observation rather than the preceding calendar day. Percentage volume changes were calculated only when the preceding retained volume was valid and non-zero. This convention allows lags to cross weekends, holidays, and year boundaries without creating artificial non-trading observations.

The retained series also enforced logical OHLC relationships before lag construction. One source row with an inconsistent OHLC range was excluded, while rows with missing or invalid trading volume were retained in the audit but not in the analytical sequence. Constructing lags only after these exclusions means that both price and volume changes refer to the immediately preceding retained observation. This implementation detail is important for reproducibility because using the previous raw row instead would create a different return and volume-change sequence whenever an excluded observation occurred.

2.4. Construction of Price and Return Variables

Let P_t denote the BRMS closing price at retained trading observation t and P_{t-1} the closing price at the immediately preceding retained trading observation.

The Daily Closing Price Change was calculated as:

$$\Delta P_t = P_t - P_{t-1} \quad (1)$$

Daily price direction was classified as Increase when $\Delta P_t > 0$, Decrease when $\Delta P_t < 0$, and Unchanged when $\Delta P_t = 0$.

The Daily Stock Return (%) was calculated as a simple close-to-close percentage price return:

$$R_t = [(P_t - P_{t-1}) \ / \ P_{t-1}] \times 100 \quad (2)$$

where R_t is the percentage close-to-close return at observation t (Campbell et al., 1997).

To characterize the magnitude of a daily price movement independently of its direction, the Absolute Daily Stock Return (%) was defined as:

$$AR_t = |R_t| \quad (3)$$

The signed return and its absolute magnitude were retained separately so that tail magnitude and direction could be distinguished (Ding et al., 1993).

Annual closing-price performance compared the first and final retained closing prices in each calendar year:

$$APC_\gamma = [(P_\gamma,T - P_{\gamma,1}) \ / \ P_{\gamma,1}] \times 100 \quad (4)$$

Thus, annual performance is a first-to-final retained closing-price comparison rather than a within-year minimum-to-maximum range.

2.5. Percentile-Based Price-Anomaly Identification

Price anomalies were identified from the distribution of Absolute Daily Stock Return (%) across the 2,389 valid paired observations. Empirical percentiles were computed by linear interpolation equivalent to the Type 7 sample-quantile convention in the Hyndman–Fan taxonomy (Hyndman & Fan, 1996). The 95th percentile was used as a study-specific upper-tail boundary; it is an operational rule for this sample, not a universal financial-market cutoff.

The resulting price threshold was 8.09529381479488%, reported as 8.0953% in the manuscript. A return above this boundary was classified as a Sharp Price Increase, a return below its negative counterpart as a Sharp Price Decrease, and all other observations as normal. Strict inequalities were used so that a value exactly equal to a threshold remained in the normal category. This rule was fixed before the joint-anomaly and contextual stages and was not adjusted to increase the number of events.

The resulting BRMS price-anomaly threshold was:

$$Q^P{}_{0\cdot 95} = 8.09529381479488\% \quad (5)$$

For presentation, the threshold is reported as 8.0953%.

The signed daily return was then classified using strict inequalities:

$$\begin{gathered} \textbf{Price Category}_{\mathbf{t}} = \\ \text{Sharp Price Increase, if } R_t > 8.09529381479488\% \\ \text{Sharp Price Decrease, if } R_t < -8.09529381479488\% \\ \text{Normal Price Movement, otherwise} \end{gathered} \quad (6)$$

Strict inequalities were used, so observations exactly equal to a boundary remained classified as normal.

2.6. Trading-Volume Variables and Volume-Anomaly Identification

Daily trading volume V_t denotes the number of BRMS shares reported as traded on observation t. Volume was analyzed both in shares and as a signed percentage change from the immediately preceding retained observation. Daily volume is a standard market-activity variable with strongly heterogeneous empirical behavior (Ajinkya & Jain, 1989).

The Daily Trading Volume Change in shares was defined as:

$$\Delta V_t = V_t - V_{t-1} \tag{7}$$

The corresponding signed percentage change was calculated as:

$$VC_t = [(V_t - V_{t-1}) \ / \ V_{t-1}] \times 100 \tag{8}$$

The percentage measure was calculated only when V_{t-1} was valid and non-zero; its absolute value was retained descriptively but was not used to set anomaly thresholds.

$$AVC_t = |VC_t| \tag{9}$$

Volume anomalies were classified from the signed VC_t distribution using the Type 7 5th and 95th percentiles. These lower- and upper-tail choices are study-specific operational rules.

The estimated signed-volume thresholds were −83.77610145136153% and 637.3560629581457%, reported as −83.7761% and 637.3561%. Values below the lower boundary were classified as Abnormally Large Volume Decreases and values above the upper boundary as Abnormally Large Volume Increases. Because the thresholds are empirical quantiles of the same retained sample, the full-sample lower- and upper-tail counts are expected to be approximately balanced even though their annual distribution can be highly uneven. The directional volume categories were retained for joint-pattern classification rather than collapsed into a single unsigned magnitude measure.

The estimated BRMS thresholds were:

$$Q^{V}{}_{0\cdot 05} = -83.77610145136153\% \tag{10}$$

$$Q^{V}{}_{0\cdot 95} = 637.3560629581457\% \tag{11}$$

Accordingly:

$$\begin{gathered} \textbf{Volume Category}_{\mathbf{t}} = \\ \text{Abnormally Large Volume Decrease, if } VC_t < -83.77610145136153\% \\ \text{Abnormally Large Volume Increase, if } VC_t > 637.3560629581457\% \\ \text{Normal Daily Volume Change, otherwise} \end{gathered} \tag{12}$$

Strict inequalities were used at both boundaries.

The signed percentage-change measure is mechanically asymmetric: declines in positive volume approach −100%, whereas positive proportional increases are unbounded. Very small preceding-volume denominators can therefore generate extremely large positive changes, so percentage anomalies were interpreted together with underlying share volumes.

2.7. Simultaneous Price–Volume Anomalies and Joint Pattern Classification

Directional price and volume categories were converted to binary anomaly indicators, A^{P}_{t} and A^{V}_{t}. A simultaneous anomaly was recorded only when both indicators equaled one on the same retained trading observation.

$A^P{}_t = 1$ if observation t is a Sharp Price Increase or Sharp Price Decrease; 0 otherwise

$A^V{}_t = 1$ if observation t is an Abnormally Large Volume Increase or Decrease; 0 otherwise

A Simultaneous Price–Volume Anomaly was defined as:

$$J_t = A^P{}_t \times A^V{}_t \tag{13}$$

such that $J_t = 1$ only when both dimensions were anomalous on the same retained trading observation.

Joint events were further classified into four directional patterns:

1. Sharp Price Increase + Large Volume Increase;
2. Sharp Price Increase + Large Volume Decrease;
3. Sharp Price Decrease + Large Volume Increase; and
4. Sharp Price Decrease + Large Volume Decrease.

The term simultaneous denotes contemporaneous classification only; it does not establish whether price or volume moved first or caused the other variable to change.

2.8. Statistical Association Between Price and Trading-Volume Anomalies

Association between binary Price-Anomaly Status and Volume-Anomaly Status was evaluated with a 2 × 2 contingency table. Pearson's chi-square test of independence was the primary inferential test, with the Yates continuity-corrected chi-square and a two-sided Fisher exact test reported as complementary assessments (Agresti, 2013; Fisher, 1922; Yates, 1934). All tests used $\alpha = 0.05$.

For the contingency table, the four observed cells correspond to both anomalies, price anomaly only, volume anomaly only, and neither anomaly. Pearson's chi-square was used as the principal test because expected counts were adequate; Yates correction and the two-sided Fisher exact test were retained as complementary checks rather than substitutes for the sample odds ratio. The odds ratio summarizes relative odds from the observed table, while Phi/Cramér's V summarizes the numerical magnitude of dependence. Reporting these measures together avoids treating a small p-value as evidence that the association is large or economically dominant.

Relative association was summarized with the sample odds ratio OR = ad/bc, reported separately from Fisher's exact significance test. A 95% log-Wald confidence interval used SE[ln(OR)] = (1/a + 1/b + 1/c + 1/d)^(1/2). Association magnitude was summarized by Phi; for a 2 × 2 table, |Phi| equals Cramér's V. Statistical significance was interpreted separately from magnitude.

$$OR = ad \,/\, bc \tag{14}$$

2.9. Event- and News-Based Contextual Validation

The association analysis was complemented by structured event- and news-based contextual validation. Its purpose was to assess whether selected simultaneous anomalies corresponded with credible, temporally eligible public information; it did not attribute causality. Events were prespecified independently of whether a plausible explanation could later be located. The selection retained the largest-absolute-return joint anomaly from each year containing joint events and all less frequent directional configurations, yielding Events 2, 4, 5, 7, 11, 12, 14, 16, 20, 21, and 23.

The primary search window was t−3 through t calendar days, allowing weekend and holiday releases to be captured. A restricted t+1 to t+2 documentation window was used only when a later source documented information that had already become public on or before t; genuinely later information was not admitted as an explanation for an earlier anomaly.

Sources were prioritized from issuer, regulatory, government, exchange, and index-provider documents to established independent financial reporting. Each event was evaluated for temporal proximity, source credibility, substantive relevance, and directional consistency. Evidence was classified as High when correspondence was direct, specific, credible, and temporally proximate; Moderate when correspondence remained indirect or ambiguous; and Not Externally Validated when sufficiently specific, credible, and temporally eligible public evidence could not be established.

The source hierarchy and fixed timing window were designed to constrain interpretive discretion. Direct issuer, regulatory, government, exchange, and index-provider evidence was preferred because these sources provide the strongest security-specific documentation. Established financial media were used when primary evidence was unavailable, when publication timing required corroboration, or when a contemporaneous market narrative was relevant. Importantly, a plausible item outside the allowable window was not admitted merely because it appeared economically attractive. This rule makes Not Externally Validated a statement about evidence obtained under the protocol, not a statement about the true absence of information.

The procedure is structured contextual validation rather than a formal event study: it does not estimate expected returns, abnormal returns, cumulative abnormal returns, or a market-model counterfactual (Kothari & Warner, 2007). Accordingly, contextual correspondence is documentary evidence, not causal event-study inference.

2.10. Computational Implementation and Reproducibility

Data preparation, variable construction, anomaly identification, and contingency analysis were implemented using the Yahoo Finance Clean Price–Volume Anomaly Calculator v2.1 within the STATCAL ONLINE analytical workflow. The Python-based application used pandas and NumPy for data transformation and percentile calculations and SciPy for inferential procedures. The analytical files preserve cleaned observations, metadata, methodology settings, exclusion audits, thresholds, lag conventions, and calculated variables. Package versions were not reconstructed because the original analytical metadata did not preserve them.

The reproducibility record also preserves the exact empirical thresholds, strict boundary rules, chronological ordering, and contingency counts used in the manuscript. These files allow the principal results to be traced from retained market observations to the reported classifications without reconstructing decisions from narrative text alone. The study nevertheless avoids reporting unverified package-version numbers, because retrospectively guessing software versions would create false precision rather than improve reproducibility.

3. Results

3.1. Closing-Price Dynamics, Daily Returns, and Price Anomalies

To provide an integrated overview of the stock-price behavior of PT Bumi Resources Minerals Tbk (BRMS.JK), Figure 3.1 presents the time-series evolution of the closing price, daily closing-price change, daily stock return, and absolute daily stock return over the 2016–2025 observation period. The four panels distinguish the long-run evolution of the price level from the direction and magnitude of shorter-term daily movements.

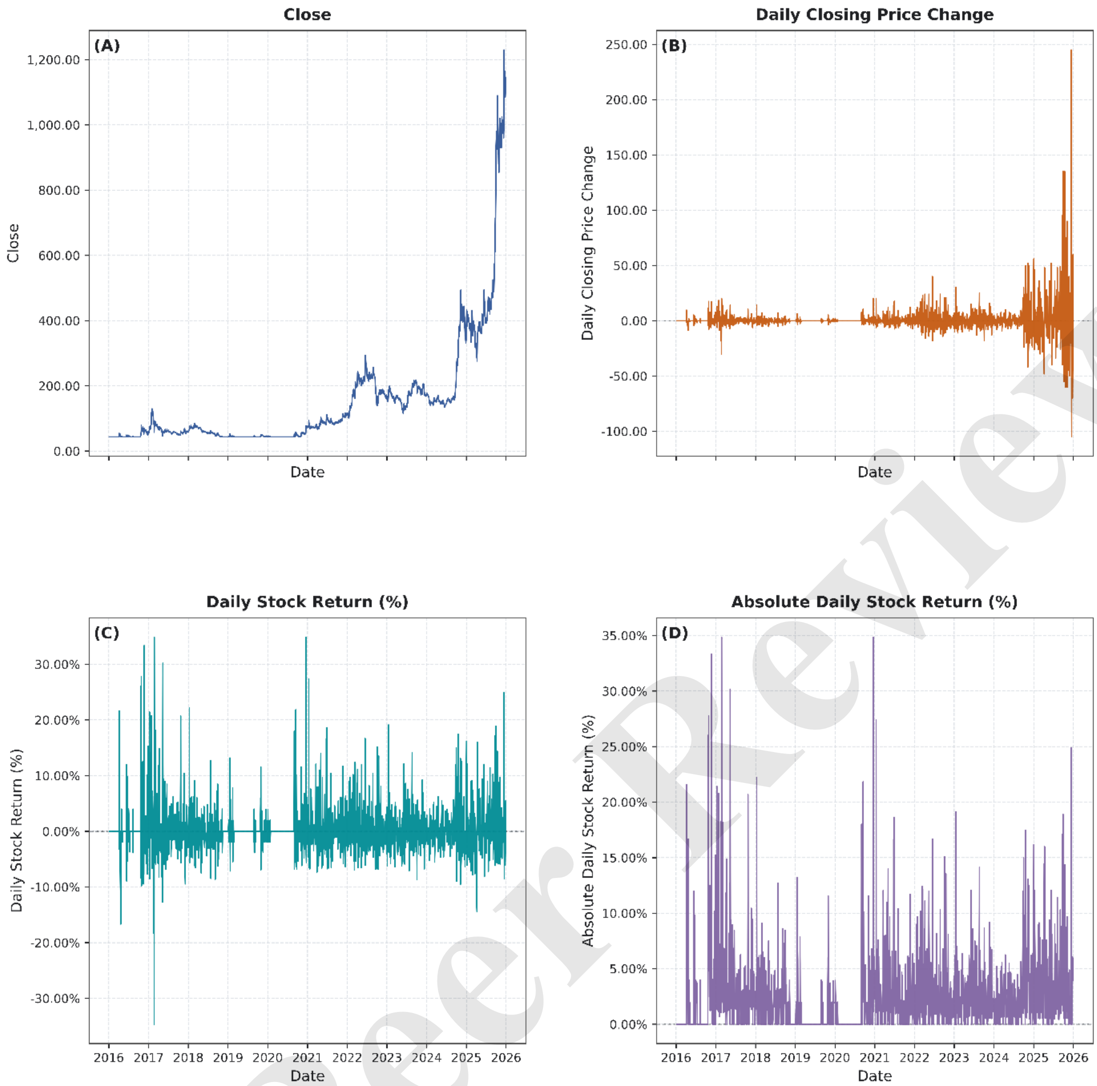

Figure 3.1 Time-Series Dynamics of Closing Price, Daily Closing-Price Change, Daily Stock Return, and Absolute Daily Stock Return of PT Bumi Resources Minerals Tbk (BRMS.JK), 2016–2025

Note. Panel A presents the daily closing price; Panel B presents the daily closing-price change; Panel C presents the signed daily stock return; and Panel D presents the absolute daily stock return.

Figure 3.1 shows a low-price regime in the early sample followed by a pronounced rise toward 2024–2025, with the sample closing-price maximum of IDR 1,230 reached in December 2025. Daily price changes and returns remained concentrated near zero for much of the period but contained isolated positive and negative extremes. Absolute returns confirm that most observations were moderate relative to a small set of large tail movements.

Table 3.1. Descriptive Statistics of Closing-Price Levels, Changes, and Returns of PT Bumi Resources Minerals Tbk (BRMS.JK), 2016–2025

Variable	Minimum	Maximum	Mean	Standard Deviation
Close	43.61	1230	150.4345	179.5334
Daily Closing Price Change	-105	245	0.4422	11.4761
Daily Stock Return (%)	-34.6464	34.8445	0.2186	4.1805
Absolute Daily Stock Return (%)	0	34.8445	2.3635	3.4549

Note. Descriptive statistics for the closing price are based on 2,390 retained trading observations. Statistics for daily closing-price change, daily stock return, and absolute daily stock return are based on 2,389 valid paired observations.

Table 3.1 summarizes the dispersion underlying the figure. Closing prices ranged from IDR 43.61 to IDR 1,230 (mean 150.4345), while daily returns ranged from −34.6464% to 34.8445% (mean 0.2186%, SD 4.1805%). The mean absolute return was 2.3635%, indicating that the largest daily moves were far larger than the typical observation.

To examine how the daily price dynamics accumulated at the annual level, Table 3.2 reports the initial and final closing prices, absolute price changes, and percentage price changes for each calendar year.

Table 3.2. Annual Closing-Price Performance and Observation Coverage of PT Bumi Resources Minerals Tbk (BRMS.JK), 2016–2025

Year	Observation Period	Initial Closing Price (IDR)	Final Closing Price (IDR)	Absolute Closing Price Change (IDR)	Percentage Closing Price Change (%)	Annual Price Movement
2016	2016-01-04 – 2016-12-30	43.61	58.44	14.83	34.01	Increase
2017	2017-01-03 – 2017-12-29	56.69	57.57	0.88	1.55	Increase
2018	2018-01-02 – 2018-12-28	58.44	43.61	-14.83	-25.38	Decrease
2019	2019-01-02 – 2019-12-30	43.61	45.35	1.74	3.99	Increase
2020	2020-01-02 – 2020-12-30	44.48	72.39	27.91	62.75	Increase
2021	2021-01-04 – 2021-12-30	75.88	116.00	40.12	52.87	Increase
2022	2022-01-03 – 2022-12-30	112.00	159.00	47.00	41.96	Increase
2023	2023-01-02 – 2023-12-29	160.00	170.00	10.00	6.25	Increase
2024	2024-01-02 – 2024-12-30	175.00	346.00	171.00	97.71	Increase
2025	2025-01-02 – 2025-12-30	402.00	1,100.00	698.00	173.63	Increase

Note. The initial and final closing prices correspond to the first and last retained trading observations within each calendar year. Percentage closing-price change is calculated as the difference between the final and initial closing prices divided by the initial closing price and multiplied by 100.

Table 3.2 shows positive first-to-final annual closing-price performance in nine of ten years. The only annual decline was −25.38% in 2018. Appreciation strengthened after 2020 and was largest in 2025, when the closing price rose from IDR 402 to IDR 1,100 (+173.63%). The December 2025 sample maximum of IDR 1,230 occurred before the final retained trading observation and therefore differs from the year-end value.

Because positive annual performance does not necessarily imply that positive daily movements occurred more frequently than negative or unchanged movements, Table 3.3 decomposes the observations into daily decreases, unchanged prices, and daily increases.

Table 3.3. Annual Frequency and Percentage Distribution of Daily Stock-Price Movement Directions of PT Bumi Resources Minerals Tbk (BRMS.JK), 2016–2025

Year	Decrease		Unchanged		Increase	
	f	%	f	%	f	%
2016	44	18.64	165	69.92	27	11.44
2017	124	52.54	34	14.41	78	33.05
2018	84	36.05	89	38.20	60	25.75
2019	30	12.61	184	77.31	24	10.08
2020	45	18.60	163	67.36	34	14.05

Year	Decrease		Unchanged		Increase	
	f	%	f	%	f	%
2021	111	45.12	40	16.26	95	38.62
2022	127	51.63	25	10.16	94	38.21
2023	120	50.21	22	9.21	97	40.59
2024	101	42.62	27	11.39	109	45.99
2025	115	48.73	15	6.36	106	44.92

Note. Percentages are calculated relative to the valid daily price-movement observations within each year. The first retained sample observation is excluded because no preceding closing price is available.

Despite the predominantly positive annual outcomes, daily decreases were the most frequent direction across the 2,389 paired observations: 901 decreases (37.71%), 764 unchanged observations (31.98%), and 724 increases (30.31%). Thus, annual appreciation reflected the magnitude and sequence of daily movements rather than simply the number of positive days.

The annual composition of daily directions varied substantially. Unchanged prices were especially prominent in 2016, 2019, and 2020, while decreases were the largest category in several later years. By 2025, unchanged observations represented only a small fraction of valid trading days. Figure 3.2 therefore adds information beyond the annual first-to-final returns: it shows that the path toward higher year-end prices became increasingly dominated by active upward and downward movements rather than by long sequences of unchanged closes.

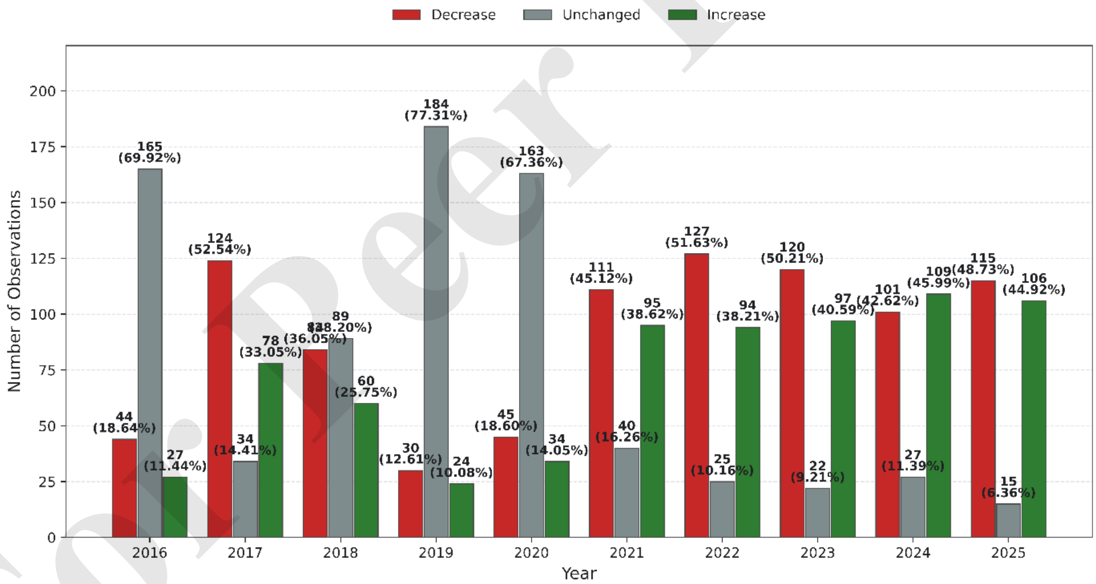

Figure 3.2 Annual Frequency and Percentage Distribution of Daily Stock-Price Movement Directions of PT Bumi Resources Minerals Tbk (BRMS.JK), 2016–2025

Note. Bar length represents frequency. Data labels report the corresponding frequency and percentage within each year.

Using the 8.0953% empirical threshold, 120 price anomalies were identified. Table 3.4 and Figure 3.3 show a strong directional imbalance toward Sharp Price Increases.

Table 3.4. Annual Frequency and Percentage Distribution of Sharp Price-Anomaly Categories of PT Bumi Resources Minerals Tbk (BRMS.JK), 2016–2025

Year	Sharp Price Increase		Sharp Price Decrease		Total	
	f	%	f	%	f	%
2016	11	61.11	7	38.89	18	100.00
2017	16	69.57	7	30.43	23	100.00
2018	4	66.67	2	33.33	6	100.00
2019	2	100.00	0	0.00	2	100.00
2020	7	100.00	0	0.00	7	100.00
2021	9	100.00	0	0.00	9	100.00
2022	14	100.00	0	0.00	14	100.00
2023	6	85.71	1	14.29	7	100.00
2024	9	81.82	2	18.18	11	100.00
2025	20	86.96	3	13.04	23	100.00

Note. Percentages are calculated relative to the total number of price anomalies identified within each year. A daily return greater than 8.0953% is classified as a Sharp Price Increase, while a return below −8.0953% is classified as a Sharp Price Decrease.

Of the 120 price anomalies, 98 (81.67%) were Sharp Price Increases and 22 (18.33%) were Sharp Price Decreases. Upward anomalies outnumbered downward anomalies in every year, and no Sharp Price Decrease was identified from 2019 through 2022. The largest yearly anomaly counts occurred in 2017 and 2025, with 23 events in each year. The ten largest absolute returns are reported in Supplementary Table S2.

The annual price-anomaly distribution was also uneven. The largest numbers of price anomalies occurred in 2017 and 2025, with 23 events in each year, but their directional composition differed: 2017 contained both sharp increases and decreases, whereas 2025 was heavily weighted toward increases. From 2019 through 2022 every identified price anomaly was upward. Figure 3.3 makes this asymmetry visible across years and confirms that the dominance of Sharp Price Increases was not confined to one isolated calendar period.

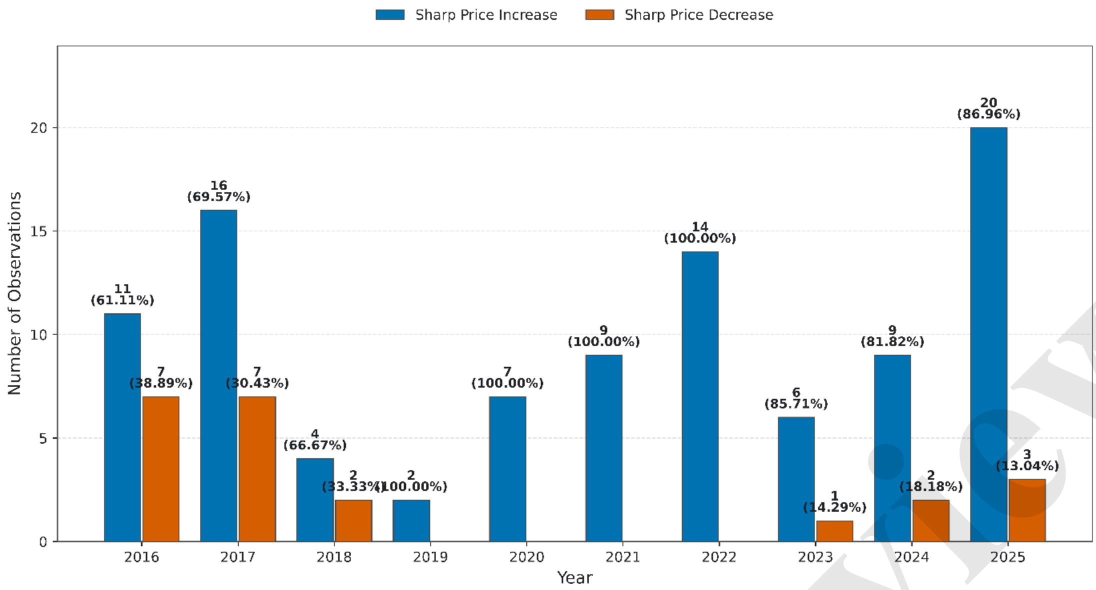

Figure 3.3 Annual Frequency and Percentage Distribution of Sharp Price-Anomaly Categories of PT Bumi Resources Minerals Tbk (BRMS.JK), 2016–2025

Note. Bar length represents the annual anomaly frequency. Data labels report the frequency and percentage of each category relative to the total price anomalies identified within the corresponding year.

3.2. Trading-Volume Dynamics and Volume Anomalies

Figure 3.4 summarizes BRMS trading-volume dynamics in levels, absolute share changes, signed percentage changes, and absolute percentage changes. Volume was highly episodic, with several multi-billion-share peaks. Percentage changes display much greater visual asymmetry because a small number of very large positive changes dominate the scale.

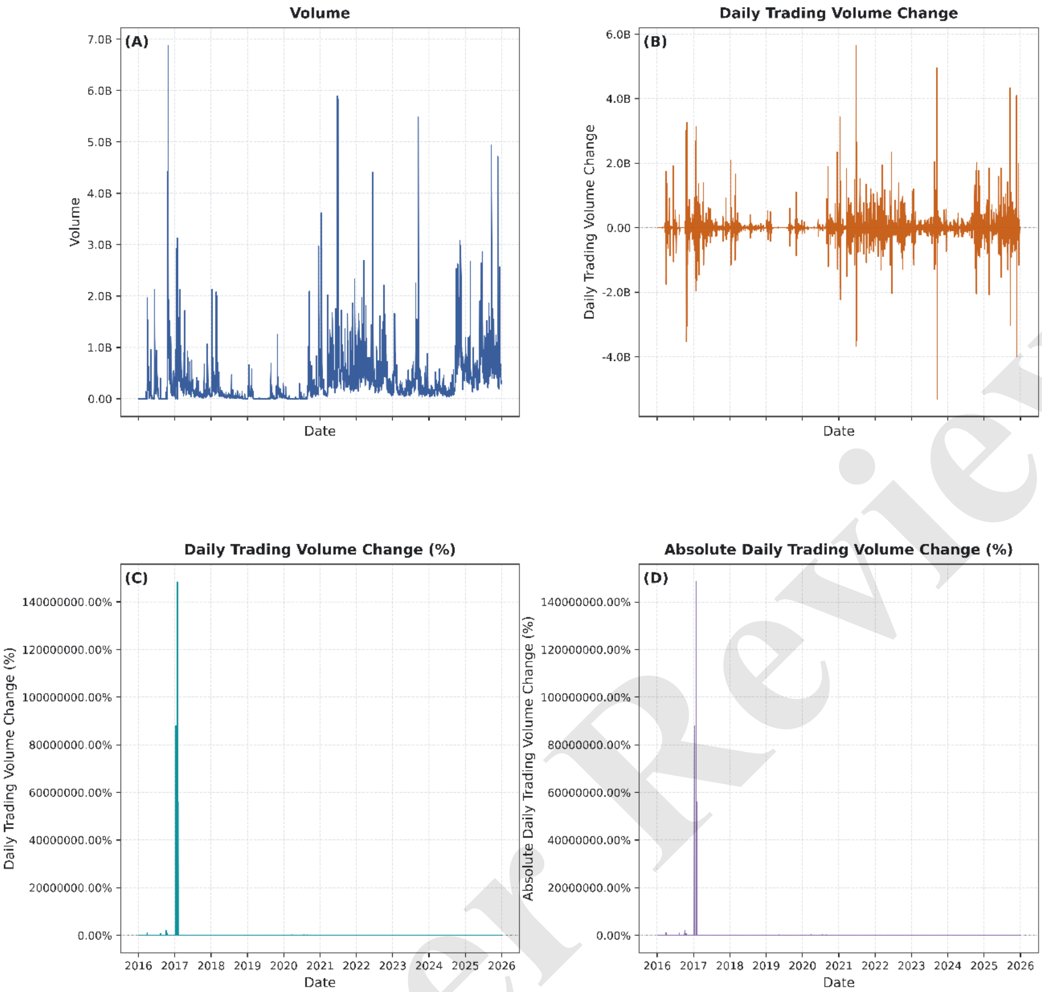

Figure 3.4 Time-Series Dynamics of Trading Volume, Daily Trading-Volume Change, Daily Trading-Volume Change (%), and Absolute Daily Trading-Volume Change (%) of PT Bumi Resources Minerals Tbk (BRMS.JK), 2016–2025

Note. Panel A presents daily trading volume; Panel B presents the change in trading volume relative to the immediately preceding retained trading observation; Panel C presents the corresponding percentage change; and Panel D presents the absolute magnitude of the daily percentage change in trading volume.

The highest retained daily volume was 6.879 billion shares, while the largest absolute increase and decrease in daily volume were approximately +5.650 billion and −5.305 billion shares. These large level changes occurred at several points in the sample rather than in a single year.

The percentage measure is especially sensitive to low preceding-volume bases. Because positive proportional increases are unbounded whereas declines approach −100%, percentage changes can appear extremely large when a very small prior volume is followed by ordinary or high trading activity.

To complement the time-series visualization and quantify the dispersion observed across the four panels, Table 3.5 reports the minimum, maximum, mean, and standard deviation of daily trading volume and the corresponding absolute and percentage changes.

Table 3.5. Descriptive Statistics of Trading Volume and Daily Trading-Volume Changes of PT Bumi Resources Minerals Tbk (BRMS.JK), 2016–2025

Variable	Minimum	Maximum	Mean	Standard Deviation
Volume	114	6879489073	363543127.3812	569400297.8313

Variable	Minimum	Maximum	Mean	Standard Deviation
Daily Trading Volume Change	-5304659900	5649823018	121270.5856	493525205.1351
Daily Trading Volume Change (%)	-99.9999	148418425.7482	124991.3269	3710664.0254
Absolute Daily Trading Volume Change (%)	0.0000	148418425.7482	125038.3035	3710662.4421

Note. Trading volume is measured in shares. Daily Trading Volume Change represents the difference between the current and immediately preceding retained trading observations. Percentage changes are calculated relative to the preceding retained trading volume.

Table 3.5 quantifies this dispersion: volume ranged from 114 to 6,879,489,073 shares, and the standard deviation of volume exceeded its mean. Signed percentage changes ranged from −99.9999% to 148,418,425.7482%. These extreme summary values are driven by denominator sensitivity and should be interpreted with the underlying share volumes; detailed rankings are provided in Supplementary Tables S3–S6.

The ranked observations in the supplement clarify why the percentage distribution is so extreme. Several of the largest positive percentage changes followed preceding volumes of only hundreds or thousands of shares and then moved to millions or billions of shares. Conversely, the largest negative percentage changes clustered close to −100% because a positive volume cannot decline by more than its entire preceding base. These mechanical features do not invalidate the measure, but they explain why its arithmetic mean and standard deviation are dominated by a small set of observations.

Table 3.6. Annual Frequency and Percentage Distribution of Daily Trading-Volume-Change Anomaly Categories of PT Bumi Resources Minerals Tbk (BRMS.JK), 2016–2025

Year	Abnormally Large Volume Increase		Abnormally Large Volume Decrease		Total	
	f	%	f	%	f	%
2016	33	46.48	38	53.52	71	100.00
2017	6	66.67	3	33.33	9	100.00
2018	9	39.13	14	60.87	23	100.00
2019	24	47.06	27	52.94	51	100.00
2020	35	50.00	35	50.00	70	100.00
2021	5	100.00	0	0.00	5	100.00
2022	1	100.00	0	0.00	1	100.00
2023	5	71.43	2	28.57	7	100.00
2024	0	0.00	0	0.00	0	0.00
2025	2	66.67	1	33.33	3	100.00

Note. Percentages are calculated relative to the total number of trading-volume-change anomalies identified within each year. The lower and upper anomaly thresholds are −83.7761% and 637.3561%, respectively. A total of zero indicates that no trading-volume-change anomaly was identified in the corresponding year.

The 5th/95th percentile rule identified 240 volume anomalies, evenly split between 120 Abnormally Large Volume Increases and 120 Abnormally Large Volume Decreases. The balance in full-sample counts follows from the two empirical tails, but the annual allocation was highly uneven.

Volume anomalies were concentrated in the first half of the sample: 224 of 240 events (93.33%) occurred during 2016–2020, including 71 in 2016, 51 in 2019, and 70 in 2020. Only 16 occurred during 2021–2025, none in 2024. Table 3.6 and Figure 3.5 therefore show strong temporal heterogeneity under the fixed full-sample thresholds, not a tested structural break.

The yearly counts also show that the balanced full-sample tails did not translate into balanced annual behavior. Abnormally Large Volume Decreases were more frequent in some early years, while increases predominated in several later years with very few total anomalies. The sharp reduction in anomaly counts after 2020 is therefore the more informative temporal result than the 120/120 full-sample split. Under the fixed thresholds, the distribution indicates that extreme proportional changes in trading activity were concentrated in the early sample even though high absolute trading volumes continued to occur later.

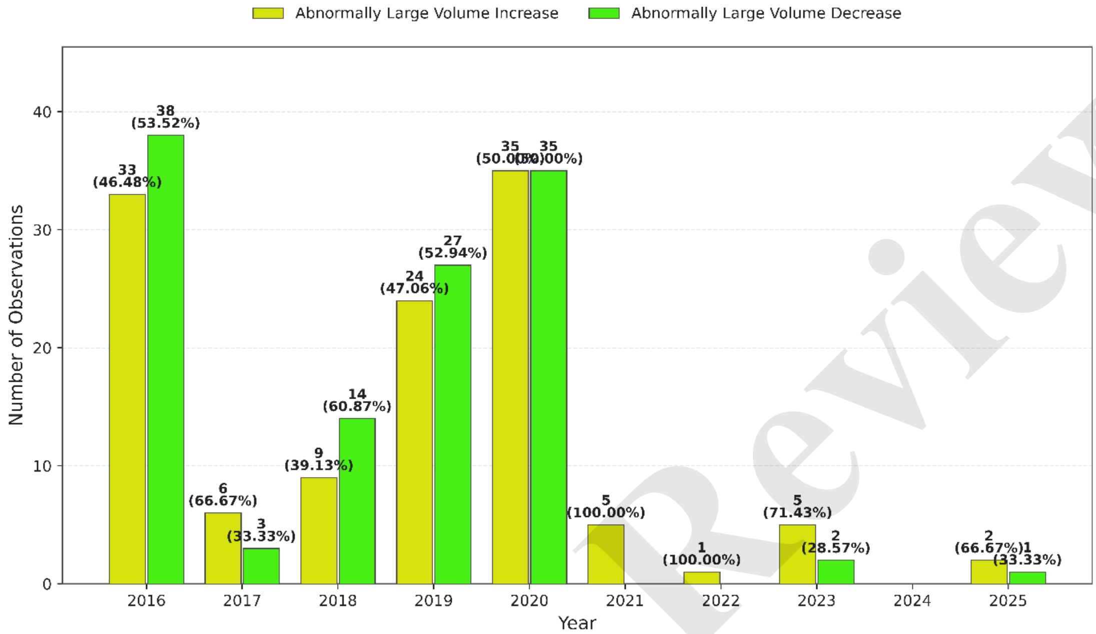

Figure 3.5 Annual Frequency and Percentage Distribution of Daily Trading-Volume-Change Anomaly Categories of PT Bumi Resources Minerals Tbk (BRMS.JK), 2016–2025

Note. Bar height represents frequency. Data labels report the frequency and percentage of each anomaly category relative to the total number of trading-volume-change anomalies identified within each year. No anomaly was identified in 2024.

3.3. Simultaneous Price–Volume Anomalies and Statistical Association

Across 2,389 valid paired observations, 23 simultaneous price-volume anomalies were identified; Supplementary Table S7 provides the complete event list.

Directional concentration was pronounced: 20 of 23 events (86.96%) combined a Sharp Price Increase with an Abnormally Large Volume Increase, two combined a price increase with a volume decrease, one combined a price decrease with a volume decrease, and none combined a price decrease with a volume increase.

The 23 joint events were distributed across 2016–2023 and 2025, with none in 2024. The largest annual count occurred in 2017, when six events also displayed the greatest diversity of directional combinations. From 2018 onward, joint anomalies were almost exclusively increase/increase observations. This temporal pattern supports the use of the full event list in the supplement while allowing the main text to focus on the dominant direction and aggregate association rather than narrating each date separately.

The joint classification denotes same-observation co-occurrence only. Table 3.7 shows that the increase/increase pattern occurred in every year with at least one joint anomaly; the greatest annual count was six events in 2017.

Table 3.7. Annual frequency of simultaneous price–volume anomaly combinations of PT Bumi Resources Minerals Tbk (BRMS.JK), 2016–2025

Year	Sharp Price Increase + Large Volume Increase	Sharp Price Increase + Large Volume Decrease	Sharp Price Decrease + Large Volume Increase	Sharp Price Decrease + Large Volume Decrease	Total Joint Anomaly Events
2016	3	0	0	1	4
2017	4	2	0	0	6
2018	1	0	0	0	1
2019	1	0	0	0	1
2020	3	0	0	0	3
2021	4	0	0	0	4
2022	1	0	0	0	1
2023	2	0	0	0	2
2024	0	0	0	0	0
2025	1	0	0	0	1

Note. The four joint patterns represent all possible combinations of Sharp Price Increase or Sharp Price Decrease with Abnormally Large Volume Increase or Abnormally Large Volume Decrease.

Overall, the directional distribution was therefore strongly asymmetric toward positive-price/high-volume observations, although joint anomalies themselves remained rare relative to the full sample.

To determine whether the two anomaly statuses co-occurred more frequently than expected under independence, Table 3.8 reports the 2 × 2 contingency results.

Table 3.8. Statistical association between price and trading-volume anomalies of PT Bumi Resources Minerals Tbk (BRMS.JK), 2016–2025

Measure	Estimate	p-value	Interpretation
Both price and volume anomalies	23		Observed joint events
Price anomaly only	97		Price anomaly without volume anomaly
Volume anomaly only	217		Volume anomaly without price anomaly
Neither anomaly	2052		No anomaly on either dimension
Valid paired observations	2389		Complete price–volume classifications
Expected simultaneous anomalies under independence	12.0553		Expected cell count
Pearson chi-square	11.6304	0.0006	Two-sided test of independence
Yates-corrected chi-square	10.5921	0.0011	Continuity-corrected 2 × 2 test
Sample odds ratio	2.2422		95% CI: 1.39–3.61
Two-sided Fisher exact test	—	0.0016	Exact test of independence
Phi coefficient / Cramer's V	0.0698		Positive association; limited association magnitude
Minimum expected cell count	12.0553		Expected-count criterion
Association direction	Positive association		Based on the odds ratio

Note. Tests are based on 2,389 observations with complete price- and trading-volume-anomaly classifications. Pearson's chi-square test is reported as the primary test of independence, with the Yates continuity correction and two-sided Fisher's exact test reported as complementary checks. The Phi coefficient/Cramér's V summarizes association magnitude. All reported p-values are two-sided.

Under independence, approximately 12.06 joint events were expected; 23 were observed, an observed-to-expected ratio of about 1.91. A volume anomaly occurred in 19.17% of price-anomaly observations versus 9.56% of non-price-anomaly observations.

The contingency counts were 23 observations with both anomalies, 97 with a price anomaly only, 217 with a volume anomaly only, and 2,052 with neither. These counts imply 120 price anomalies and 240 volume anomalies in the valid paired sample. Simultaneous events represented only about 0.96% of all paired observations, which is why the statistically significant dependence should be interpreted alongside the small effect-size measure rather than as a pervasive daily relationship.

Pearson's chi-square test rejected independence ($\chi^2 = 11.6304$, $p = 0.0006$), with a minimum expected cell count of 12.0553. The Yates-corrected result was also significant ($\chi^2 = 10.5921$, $p = 0.0011$).

The two-sided Fisher exact test was significant ($p = 0.0016$). Separately, the sample odds ratio was 2.2422 (95% CI: 1.39–3.61), meaning that the odds of a volume anomaly were approximately 2.24 times as high among price-anomaly observations as among non-price-anomaly observations.

Phi/Cramér's V was 0.0698, indicating limited association magnitude despite statistically detectable dependence. The result therefore supports positive co-occurrence without implying a strong, deterministic, predictive, or causal relationship.

3.4. Event- and News-Based Contextual Validation of Selected Joint Anomalies

Structured contextual validation was applied to 11 prespecified events (2, 4, 5, 7, 11, 12, 14, 16, 20, 21, and 23) using the fixed t−3 to t calendar-day window and the restricted t+1 to t+2 documentation rule. Table 3.9 summarizes the classifications; Supplementary Table S8 preserves the full source and timing audit.

Table 3.9. Summary of Event- and News-Based Contextual Validation of Selected Joint Anomalies

Event	Date	Return (%)	Volume change (%)	Evidence strength
2	20 Apr 2016	−16.66	−99.00	Not Externally Validated
4	19 Oct 2016	+26.00	+795,118.79	Moderate
5	9 Jan 2017	+21.44	≈−100.00	Not Externally Validated
7	26 Jan 2017	+20.80	≈−100.00	Moderate
11	11 Jan 2018	+22.21	+4,062.96	Moderate
12	30 Oct 2019	+11.55	+747.72	High
14	15 Sep 2020	+21.83	+950.68	Not Externally Validated
16	13 Jan 2021	+27.38	+1,848.93	Not Externally Validated
20	21 Jan 2022	+9.01	+758.75	Not Externally Validated
21	16 Jan 2023	+19.11	+897.52	Moderate
23	19 Sep 2025	+17.12	+720.72	High

Note. Detailed contextual evidence, source citations, and timing-audit information are provided in Supplementary Table S8. Contextual correspondence is descriptive and does not establish causal attribution.

Two events received High classifications. On 30 October 2019, same-day reporting documented positive 9M 2019 financial results and expected progress at the Poboya gold operation, corresponding with an 11.55% price increase and 747.72% volume increase (PT Bumi Resources Minerals Tbk, 2019; Sulmaihati, 2019).

The second High event occurred on 19 September 2025. VanEck's 17 September documentation identified BRMS as a new pro forma MarketVector Global Gold Miners Index constituent at a 0.61% weight; the implementation date coincided with a 17.12% price increase and a 720.72% volume increase (Putri, 2025; Tim Riset IDX Channel, 2025; VanEck, 2025). This is strong temporal and security-specific correspondence, not proof that index inclusion was the sole cause of the movement.

Four events were classified Moderate because the available evidence was indirect, group-level, market-wide, temporally less precise, or internally inconsistent with one dimension of the fixed anomaly classification. The audit retained such ambiguity rather than upgrading events on the basis of a plausible narrative.

The Moderate category captured several different forms of incomplete correspondence. Some cases involved Bakrie-group or broad market information that was directionally consistent with the BRMS price movement but not sufficiently issuer-specific. One January 2017 event also contained tension between external descriptions of active trading and the fixed Yahoo-based classification of a large volume decrease. Retaining that inconsistency in the audit was preferable to forcing the event into a stronger category and illustrates why source credibility and directional consistency were evaluated separately.

Five events were Not Externally Validated because sufficiently specific, credible, and temporally eligible public information could not be established under the protocol. Some plausible corporate information fell outside the fixed window and was therefore excluded. This classification does not imply that no information existed or that undocumented mechanisms caused the events.

Three of the Not Externally Validated cases had economically relevant information near the event date but outside the prespecified window. The 2020 first-half result, 2021 rights-issue information, and 2022 rights-issue completion were therefore excluded from the qualifying evidence set despite their apparent relevance. This timing discipline is central to the contextual design because expanding the window after observing the anomaly would increase the risk of post hoc explanation.

The final distribution—2 High, 4 Moderate, and 5 Not Externally Validated—shows that aggregate statistical dependence and event-level documentary correspondence are distinct evidence layers. Neither layer, separately or jointly, establishes causal attribution.

4. Discussion

The results are best interpreted as complementary layers rather than a single mechanism. Long-horizon performance describes net price evolution; tail classifications identify unusual observations; contingency analysis tests whether binary anomaly statuses depart from independence; and documentary validation evaluates public-information correspondence for selected events. Keeping these layers separate is essential because statistical co-occurrence does not explain an event, and documentary correspondence does not estimate a counterfactual causal effect.

4.1. Long-Term Price Performance and Daily Return Dynamics

BRMS displayed a notable contrast between long-horizon performance and daily directional frequency. Nine of ten years ended with a higher closing price, yet daily decreases were more frequent than daily increases. This is consistent with the arithmetic of cumulative performance: a smaller number of sufficiently large gains can outweigh a larger number of smaller losses. The finding therefore illustrates why sign frequency, return magnitude, and horizon should not be treated as interchangeable measures of market performance.

This distinction has a direct interpretive implication for empirical studies of long-horizon stock performance. A positive annual endpoint does not indicate that investors encountered predominantly positive daily movements, just as a high frequency of negative days does not imply a negative annual return. For BRMS, the coexistence of nine positive annual outcomes with more frequent daily declines highlights the role of return size and sequence. The result therefore supports reporting both horizon-level performance and daily distributional behavior when a study is concerned with unusual price movements.

The presence of several large absolute returns reinforces the importance of tail analysis. Unconditional averages compress these observations, whereas the empirical-tail framework makes their direction and timing explicit. The strong appreciation in 2024–2025 remains descriptive, however; the study does not test momentum, sentiment, commodity exposure, or structural regime changes.

4.2. Asymmetry and Temporal Distribution of Price and Trading-Volume Anomalies

Price anomalies were strongly asymmetric: 81.67% were Sharp Price Increases. This security-specific pattern is consistent with the broader observation that positive and negative extreme returns can exhibit different behavior (Ciner, 2015; Larson & Madura, 2003; Lin & González-Rivera, 2019), but it should not be generalized to Indonesian mining stocks as a class. Tail composition depends on the security, sample, market environment, and operational definition.

The asymmetry is also consistent with the descriptive path of the security, which experienced several episodes of sharp appreciation. Even so, the percentile rule is endogenous to the BRMS sample: a different observation window would change the empirical return distribution and therefore the anomaly boundary. The finding should consequently be described as a property of this retained sample rather than evidence that upward extremes are generally more common in mining equities or in the Indonesian market.

The volume results show a different type of heterogeneity. Although lower- and upper-tail anomaly counts were balanced over the full sample, 93.33% of volume anomalies occurred during 2016–2020. This concentration is compatible with the highly heterogeneous nature of trading activity documented in prior work (Ajinkya & Jain, 1989; Gervais et al., 2001), but the present design does not identify a structural break or liquidity regime.

Interpretation of the volume tail must also account for denominator sensitivity. The most extreme positive percentage changes frequently followed exceptionally small preceding volumes; positive changes are unbounded, whereas declines approach −100%. Accordingly, the percentage metric identifies proportional extremes but should be read together with underlying share volumes rather than as a direct measure of liquidity, depth, or economic importance.

This issue illustrates a broader measurement point. Percentage transformations are useful because they place volume changes on a relative scale, but they can amplify low-base observations in a way that raw share changes do not. The study addresses this by preserving both metrics and by moving ranked underlying-volume observations to the supplement. Future applications may compare this signed percentage rule with rolling relative-volume or standardized alternatives, but those extensions would change the anomaly definition and therefore answer a different robustness question.

4.3. Joint Price-Volume Anomalies and Statistical Association

The joint-anomaly results provide the central link between price and trading activity. Twenty of the 23 joint events were increase/increase observations, a directional concentration broadly consistent with classic evidence that price changes and volume can be positively related (Campbell et al., 1993; Gallant et al., 1992; Karpoff, 1987). Yet only 23 of 2,389 observations were joint anomalies, so the pattern was concentrated in direction but rare in frequency.

The contingency analysis shows that this co-occurrence exceeded what statistical independence would imply: 23 joint events were observed versus about 12.06 expected. Pearson, Yates-corrected, and Fisher tests all indicated dependence. The odds ratio of 2.24 summarizes relative odds, not probability, and does not establish whether volume led price or price led volume.

The observed-to-expected ratio of about 1.91 is useful as a descriptive complement to the inferential tests because it makes the excess co-occurrence directly interpretable. However, it should not be read as a probability multiplier. Similarly, the odds ratio compares odds across anomaly-status groups and is not equivalent to a risk ratio. These distinctions matter because translating the estimate into 'times more likely' would overstate what the 2×2 table establishes.

Magnitude remains crucial. Phi/Cramér's V of approximately 0.070 indicates limited association magnitude even though the departure from independence was statistically detectable. This distinction follows the broader principle that statistical significance is not a measure of effect size or practical importance (Wasserstein & Lazar, 2016). The evidence therefore supports a positive but limited contemporaneous association, not a strong or predictive relationship.

The limited value of V also places the 20 increase/increase events in perspective. Directional concentration among the joint events is visually striking, but the binary association across all 2,389 observations remains small in numerical magnitude. A coherent interpretation therefore requires both statements: joint tail events occurred more often than independence predicts and were usually positive-price/high-volume observations, yet the overall dependence between the anomaly indicators was limited.

4.4. Contextual Evidence and Event-Level Heterogeneity

The contextual results reinforce the distinction between aggregate dependence and event-level explanation. Only 2 of 11 prespecified events were classified High, while 4 were Moderate and 5 were Not Externally Validated. This heterogeneity is conceptually consistent with evidence that extreme returns may or may not coincide with readily identifiable public information (Larson & Madura, 2003). It also cautions against assuming that every statistically unusual observation has a single observable catalyst.

The two High cases also demonstrate that strong documentary correspondence can arise through different information channels. The 2019 case involved issuer financial and operational information, whereas the 2025 case involved a security-specific index transition. Their common feature is not the type of information but the combination of issuer relevance, temporal eligibility, source credibility, and directional consistency. This supports the use of criteria-based validation rather than restricting the contextual stage to one predetermined class of news.

The 2019 High event involved issuer-specific financial and operational information contemporaneous with the anomaly. The relevance of accounting information to market activity is well established (Ball & Brown, 1968; Beaver, 1968), but those studies do not establish causality for BRMS. Here, the evidence supports contemporaneous correspondence only.

The 2025 High event involved pre-event documentation of BRMS inclusion in a gold-mining index. Index-composition changes can be associated with price and volume effects (Vespro, 2006), although the index effect has weakened in more recent evidence (Greenwood & Sammon, 2025). The balanced interpretation is therefore strong documentary correspondence with a security-specific index event, not proof of a mechanically determined price or flow effect.

Moderate and Not Externally Validated cases are equally informative because they preserve uncertainty. The fixed timing rule prevented plausible but out-of-window information from being admitted ex post. Not Externally Validated consequently means that qualifying public evidence could not be established under the protocol; it is not evidence of private information, manipulation, sentiment, liquidity shocks, or the absence of any information event.

The large share of non-High classifications is substantively important because it prevents the aggregate association from being converted into a narrative that every joint anomaly had a readily observable public explanation. Public archives are incomplete, and the protocol cannot rule out information that was unavailable, poorly preserved, or disseminated through channels not captured by the search. The appropriate conclusion is therefore evidence heterogeneity, not evidence absence.

The central synthesis is that statistical dependence and documentary correspondence are non-equivalent forms of evidence. BRMS anomaly statuses co-occurred more often than expected under independence, but the association magnitude was limited and strong documentary correspondence was identified for only a minority of audited events. A uniform information mechanism is therefore not supported by the design.

4.5. Empirical and Practical Implications

For empirical finance, the framework shows the value of separating tail identification, joint occurrence, association magnitude, and documentary context. These layers answer different questions and reduce the risk of treating an anomaly label as an explanation. For investors and analysts, joint price-volume anomalies may be useful screening observations that motivate closer review of disclosures, index changes, sector conditions, or market-wide information; the study does not show that they predict subsequent returns.

This separation may be especially useful in emerging-market settings where market data and historical disclosure archives can differ in availability and quality across time. A reproducible anomaly screen can identify candidate dates consistently, after which documentary review can be conducted under an explicit source and timing protocol. The sequence does not solve identification problems, but it makes clear which conclusions arise from the market data and which arise from external documentary evidence.

For market screening or surveillance, the same logic applies: unusual price-volume co-occurrence can identify observations warranting review, but it is not evidence of manipulation, insider trading, misconduct, or a regulatory violation. The practical contribution is therefore procedural and diagnostic rather than predictive or accusatory.

For practitioners, the framework is therefore most defensible as a prioritization device. A joint anomaly may justify closer examination because two market dimensions are simultaneously unusual, while the subsequent contextual stage can indicate whether a credible public-information correspondence is readily observable. Neither step determines whether the movement was justified, profitable to trade, or associated with misconduct. Those questions require different data and research designs.

5. Limitations and Future Research

Several limitations bound the inference. The study examines one security over 2016–2025, so the observed directional asymmetry, temporal concentration, joint patterns, and contextual outcomes are security- and sample-specific. The percentile thresholds are also empirical operational rules: different securities, windows, or anomaly definitions can generate different event sets. Daily data and close-to-close returns do not reveal intraday sequencing or explicit total shareholder return.

The percentage-based volume measure is sensitive to the immediately preceding trading-volume denominator; very small prior volumes can create extremely large positive percentage changes. Underlying share volumes were therefore retained for interpretation, but the measure should not be read as a direct proxy for liquidity or market depth.

Yahoo Finance served as a secondary historical-data interface. Although the workflow preserves cleaning rules, exclusions, chronology, thresholds, and audit files, future replication could compare the results with other verified market-data sources. Documentary validation is also constrained by historical page availability, timestamp quality, and archival completeness, so Not Externally Validated does not mean that no relevant information existed.

Finally, the statistical and contextual components are noncausal by design. They do not estimate lead-lag effects, expected or abnormal returns, cumulative abnormal returns, or a market-model counterfactual. Future research could apply the framework across Indonesian mining securities, use intraday data, compare alternative tail or relative-volume definitions, and conduct formal market- or sector-adjusted event studies. These extensions address different questions rather than correcting the present descriptive-associational design.

6. Conclusions

This study examined BRMS daily price and trading-volume dynamics over 2016–2025 by integrating long-horizon characterization, empirical tail classification, same-observation joint-anomaly mapping, contingency-based association testing, and prespecified documentary validation. The contribution lies in combining established analytical components into a transparent issuer-level analysis from an emerging capital market while maintaining a strict distinction between statistical dependence and causal interpretation.

The empirical tails were asymmetric across direction and time. Of 120 price anomalies, 98 were Sharp Price Increases, while 224 of 240 volume anomalies occurred during 2016–2020. Twenty-three simultaneous anomalies were identified, including 20 increase/increase events. The observed joint count exceeded the approximately 12.06 expected under independence, and the sample odds ratio was 2.24 (95% CI: 1.39–3.61), but Phi/Cramér's V of approximately 0.070 indicated limited association magnitude.

Documentary validation produced 2 High, 4 Moderate, and 5 Not Externally Validated classifications. Thus, aggregate co-occurrence and event-specific public-information correspondence represent distinct evidence layers. Joint price-volume anomalies are most appropriately treated as reproducible screening observations for further investigation, not as standalone evidence of prediction, market misconduct, or a common causal mechanism.

Ethics Statement

Ethical approval and informed consent were not applicable to the present study because the analysis relied exclusively on publicly available secondary financial-market data and publicly accessible documentary sources. The study involved no human participants, animals, surveys, interviews, experimental interventions, or personally identifiable or sensitive individual-level information.

Author contributions

Abdillah Arif Nasution: Conceptualization, Formal analysis, Writing – original draft, Writing – review & editing, Supervision, Project administration. Aulia Arif Nasution: Data curation, Validation. Prana Ugiana Gio: Methodology, Software, Investigation, Visualization.

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Disclosure statement

The authors report there are no competing interests to declare.

Data Availability Statement

The underlying historical market data used in this study are publicly accessible through Yahoo Finance. The archived MHTML source snapshot used for the analysis, cleaned analytical workbook, derived tables, calculation outputs, and computational audit files are available from the corresponding author upon reasonable request.

Declaration of generative AI use

During the preparation of this manuscript, the authors used OpenAI's ChatGPT to assist with language editing, sentence restructuring, academic readability, structural organization, and drafting support for selected manuscript sections. The tool was not used autonomously to collect or modify the underlying financial-market data, execute the statistical analyses, determine the anomaly classifications, or make final scientific decisions. All AI-assisted text, interpretations, methodological descriptions, and cited sources were critically reviewed, independently checked where appropriate, and revised by the authors. The authors take full responsibility for the accuracy, integrity, originality, interpretation, and final content of the manuscript.

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Supplementary Material

Joint Price–Volume Anomalies in an Indonesian Mining Equity: Statistical Association and Contextual Evidence from BRMS.JK, 2016–2025

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Detailed operational definitions, ranked observations, the complete simultaneous-event listing, and the full contextual-validation audit are provided here to keep the main manuscript concise.

Table S1. Operational definitions and measurement rules

Variable	Operational Definition	Measurement or Classification Rule	Scale
Closing Price (P_t)	BRMS closing price at retained trading observation t	Yahoo Finance Close field; Adjusted Close not substituted	Ratio; IDR
Initial Annual Closing Price	First retained closing price within calendar year y	$P_{\gamma,1}$	Ratio; IDR
Final Annual Closing Price	Last retained closing price within calendar year y	P_{γ},T	Ratio; IDR
Annual Closing-Price Change	Difference between final and initial retained annual closing prices	P_{γ},T − $P_{\gamma,1}$	Difference; IDR
Percentage Annual Closing-Price Change	First-to-final proportional annual closing-price movement	[(P_{γ},T − $P_{\gamma,1}$) / $P_{\gamma,1}$] × 100	Ratio-derived; %
Annual Price Movement	Direction of annual first-to-final change	Increase if >0; Decrease if <0; Unchanged if =0	Nominal
Daily Closing Price Change (ΔP_t)	Change from immediately preceding retained closing price	$P_t - P_{t-1}$	Difference; IDR
Daily Stock Return (R_t)	Simple close-to-close percentage price return	$[(P_t - P_{t-1}) / P_{t-1}] \times 100$	Ratio-derived; %
Absolute Daily Stock Return (AR_t)	Magnitude of daily return irrespective of direction	$\lvert R_t \rvert$	Ratio-derived; %
Daily Price Movement Direction	Direction of daily closing-price change	Increase; Decrease; Unchanged	Nominal
Price-Anomaly Threshold	Empirical upper-tail cutoff of absolute daily returns	Type 7 95th percentile = 8.09529381479488%	Ratio-derived; %
Price Anomaly Category	Directional extreme-return classification	Sharp Price Increase if R_t > 8.09529381479488%; Sharp Price Decrease if R_t < −8.09529381479488%; otherwise Normal	Nominal
Daily Trading Volume (V_t)	Number of BRMS shares traded at retained observation t	Historical Volume field	Ratio; shares
Daily Trading Volume Change (ΔV_t)	Change from immediately preceding retained trading volume	$V_t - V_{t-1}$	Difference; shares
Daily Trading Volume Change (%) (VC_t)	Signed proportional change in trading volume	$[(V_t - V_{t-1}) / V_{t-1}] \times 100$	Ratio-derived; %
Absolute Daily Trading Volume Change (%)	Magnitude of signed percentage volume change	$\lvert VC_t \rvert$	Ratio-derived; %

Variable	Operational Definition	Measurement or Classification Rule	Scale
Lower Volume-Anomaly Threshold	Empirical lower tail of signed volume changes	Type 7 5th percentile = −83.77610145136153%	Ratio-derived; %
Upper Volume-Anomaly Threshold	Empirical upper tail of signed volume changes	Type 7 95th percentile = 637.3560629581457%	Ratio-derived; %
Trading-Volume Anomaly Category	Directional extreme-volume classification	Large Volume Decrease if $VC_t < -83.77610145136153\%$; Large Volume Increase if $VC_t > 637.3560629581457\%$; otherwise Normal	Nominal
Price-Anomaly Status	Binary price-anomaly indicator	1 = Sharp Price Increase/Decrease; 0 = Normal	Binary
Volume-Anomaly Status	Binary volume-anomaly indicator	1 = Abnormally Large Volume Increase/Decrease; 0 = Normal	Binary
Simultaneous Price–Volume Anomaly	Price and volume both anomalous at the same retained observation	$A^P_t = 1$ and $A^V_t = 1$	Binary
Joint Price–Volume Anomaly Pattern	Directional configuration of a simultaneous anomaly	Increase–Increase; Increase–Decrease; Decrease–Increase; Decrease–Decrease	Nominal
Statistical Association	Departure of binary price- and volume-anomaly statuses from independence	2 × 2 table; Pearson χ^2; Yates correction; two-sided Fisher exact test; sample OR with 95% log-Wald CI; Phi/Cramér's V; $\alpha = .05$	Inferential
Identified Event or News Context	Credible publicly available information surrounding a selected joint anomaly	Prespecified t−3 to t search; restricted t+1 to t+2 documentation	Qualitative
Contextual Correspondence	Degree to which identified public information corresponds with the observed anomaly	Temporal proximity, source credibility, substantive relevance, directional consistency	Qualitative assessment
Evidence Strength	Overall strength of contextual evidence	High; Moderate; Not Externally Validated	Ordinal categorical
Evidence Source	Documentary evidence supporting or delimiting the contextual assessment	Primary/official sources prioritized, followed by established independent financial sources	Provenance / nominal

Note. The exact thresholds are empirical values calculated from the retained BRMS dataset. Percentiles use linear interpolation equivalent to the Type 7 convention; strict inequalities are applied.

Table S2. Top 10 largest absolute daily stock returns, 2016–2025

Rank	Year	Previous Trading Date	Previous Closing Price (IDR)	Event Trading Date	Event Closing Price (IDR)	Daily Stock Return (%)	Absolute Daily Stock Return (%)	Price Anomaly Category
1	2017	2017-02-21	57.57	2017-02-22	77.63	34.84	34.84	Sharp Price Increase
2	2020	2020-12-16	57.57	2020-12-17	77.63	34.84	34.84	Sharp Price Increase
3	2017	2017-02-20	88.09	2017-02-21	57.57	-34.65	34.65	Sharp Price Decrease
4	2016	2016-11-18	52.33	2016-11-21	69.78	33.35	33.35	Sharp Price Increase

Rank	Year	Previous Trading Date	Previous Closing Price (IDR)	Event Trading Date	Event Closing Price (IDR)	Daily Stock Return (%)	Absolute Daily Stock Return (%)	Price Anomaly Category
5	2017	2017-05-10	46.23	2017-05-12	60.18	30.18	30.18	Sharp Price Increase
6	2016	2016-10-25	62.80	2016-10-26	80.24	27.77	27.77	Sharp Price Increase
7	2021	2021-01-12	73.27	2021-01-13	93.33	27.38	27.38	Sharp Price Increase
8	2016	2016-10-18	43.61	2016-10-19	54.95	26.00	26.00	Sharp Price Increase
9	2025	2025-12-11	985.00	2025-12-12	1,230.00	24.87	24.87	Sharp Price Increase
10	2018	2018-01-10	62.80	2018-01-11	76.75	22.21	22.21	Sharp Price Increase

Note. Observations are ranked by absolute daily stock return, irrespective of sign.

Table S3. Top 10 highest daily trading volumes, 2016–2025

Rank	Year	Trading Date	Trading Volume (Shares)
1	2016	2016-10-26	6,879,489,073
2	2021	2021-06-23	5,890,471,292
3	2021	2021-06-30	5,836,848,979
4	2023	2023-09-15	5,488,746,000
5	2025	2025-09-19	4,942,216,800
6	2025	2025-11-24	4,715,327,500
7	2016	2016-10-27	4,514,733,459
8	2016	2016-10-20	4,421,268,160
9	2022	2022-06-15	4,407,775,400
10	2021	2021-06-29	3,969,056,763

Table S4. Top 10 lowest daily trading volumes, 2016–2025

Rank	Year	Trading Date	Trading Volume (Shares)
1	2016	2016-01-12	114
2	2016	2016-01-13	114
3	2016	2016-02-02	114
4	2016	2016-02-03	114
5	2016	2016-09-26	114
6	2020	2020-03-19	229
7	2016	2016-09-20	343
8	2016	2016-01-05	458
9	2016	2016-02-09	458
10	2019	2019-05-09	458

Table S5. Top 10 largest percentage increases in daily trading volume, 2016–2025

Rank	Year	Previous Trading Date	Previous Trading Volume (Shares)	Event Trading Date	Event Trading Volume (Shares)	Daily Trading Volume Change (%)
1	2017	2017-01-26	2,105	2017-01-27	3,124,209,967	148,418,425.75
2	2017	2017-01-09	1,059	2017-01-10	931,728,800	87,981,845.23
3	2017	2017-02-01	1,719	2017-02-02	963,620,651	56,056,947.76
4	2016	2016-10-04	687	2016-10-05	13,648,363	1,986,561.28
5	2016	2016-03-28	26,599	2016-03-29	287,466,465	1,080,641.63
6	2016	2016-08-09	49,758	2016-08-10	404,047,948	811,926.10
7	2016	2016-10-18	379,038	2016-10-19	3,014,181,400	795,118.79
8	2020	2020-07-21	44,140	2020-07-22	91,867,398	208,027.32
9	2020	2020-03-26	2,063	2020-03-27	3,484,264	168,793.07
10	2020	2020-08-26	100,090	2020-08-27	146,053,495	145,822.17

Table S6. Top 10 largest percentage decreases in daily trading volume, 2016–2025

Rank	Year	Previous Trading Date	Previous Trading Volume (Shares)	Event Trading Date	Event Trading Volume (Shares)	Daily Trading Volume Change (%)
1	2017	2017-01-31	1,637,307,075	2017-02-01	1,719	-100.00
2	2017	2017-01-25	1,959,644,615	2017-01-26	2,105	-100.00
3	2017	2017-01-06	536,711,619	2017-01-09	1,059	-100.00
4	2018	2018-12-19	3,003,529	2018-12-20	2,751	-99.91
5	2020	2020-03-18	232,284	2020-03-19	229	-99.90
6	2020	2020-05-13	3,808,957	2020-05-14	7,567	-99.80
7	2020	2020-02-19	36,566,543	2020-02-20	81,288	-99.78
8	2019	2019-05-08	146,180	2019-05-09	458	-99.69
9	2016	2016-03-17	139,215,899	2016-03-18	805,083	-99.42
10	2020	2020-06-11	6,901,227	2020-06-12	43,911	-99.36

Table S7. Complete simultaneous price–volume anomaly event list, 2016–2025

Year	Previous Price Trading Date	Previous Closing Price (IDR)	Event Trading Date	Event Closing Price (IDR)	Daily Stock Return (%)	Price Anomaly Category	Previous Volume Trading Date	Previous Trading Volume (Shares)	Event Trading Volume (Shares)	Daily Trading Volume Change (%)	Trading Volume Anomaly Category – Daily Change	Joint Pr Volume Anomal Pattern

Year	Previous Price Trading Date	Previous Closing Price (IDR)	Event Trading Date	Event Closing Price (IDR)	Daily Stock Return (%)	Price Anomaly Category	Previous Volume Trading Date	Previous Trading Volume (Shares)	Event Trading Volume (Shares)	Daily Trading Volume Change (%)	Trading Volume Anomaly Category – Daily Change	Joint Pr Volume Anomal Pattern
2016	2016-04-05	44.48	2016-04-06	54.08	21.58	Sharp Price Increase	2016-04-05	164,214,319	1,540,007,251	837.80	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2016	2016-04-12	52.33	2016-04-20	43.61	-16.66	Sharp Price Decrease	2016-04-12	634,085,852	6,332,555	-99.00	Abnormally Large Volume Decrease	Sharp Pr Decrease Large V Decrease
2016	2016-06-09	43.61	2016-06-10	48.84	11.99	Sharp Price Increase	2016-06-09	211,910,100	2,124,642,808	902.62	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2016	2016-10-18	43.61	2016-10-19	54.95	26.00	Sharp Price Increase	2016-10-18	379,038	3,014,181,400	795,118.79	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2017	2017-01-06	61.05	2017-01-09	74.14	21.44	Sharp Price Increase	2017-01-06	536,711,619	1,059	-100.00	Abnormally Large Volume Decrease	Sharp Pr Increase Large V Decrease
2017	2017-01-19	68.03	2017-01-20	77.63	14.11	Sharp Price Increase	2017-01-19	245,514,847	2,923,454,327	1,090.74	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2017	2017-01-25	88.09	2017-01-26	106.41	20.80	Sharp Price Increase	2017-01-25	1,959,644,615	2,105	-100.00	Abnormally Large Volume Decrease	Sharp Pr Increase Large V Decrease
2017	2017-01-26	106.41	2017-01-27	116.88	9.84	Sharp Price Increase	2017-01-26	2,105	3,124,209,967	148,418,425.75	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2017	2017-04-03	66.29	2017-04-04	74.14	11.84	Sharp Price Increase	2017-04-03	101,048,589	899,232,161	789.90	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2017	2017-10-20	50.59	2017-10-23	61.05	20.68	Sharp Price Increase	2017-10-20	36,903,848	394,204,529	968.19	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2018	2018-01-10	62.80	2018-01-11	76.75	22.21	Sharp Price Increase	2018-01-10	51,211,115	2,131,897,736	4,062.96	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2019	2019-10-29	45.35	2019-10-30	50.59	11.55	Sharp Price Increase	2019-10-29	147,934,241	1,254,066,684	747.72	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase

Year	Previous Price Trading Date	Previous Closing Price (IDR)	Event Trading Date	Event Closing Price (IDR)	Daily Stock Return (%)	Price Anomaly Category	Previous Volume Trading Date	Previous Trading Volume (Shares)	Event Trading Volume (Shares)	Daily Trading Volume Change (%)	Trading Volume Anomaly Category – Daily Change	Joint Pr Volume Anomal Pattern
2020	2020-08-31	43.61	2020-09-01	51.46	18.00	Sharp Price Increase	2020-08-31	65,893,751	965,536,710	1,365.29	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2020	2020-09-14	47.97	2020-09-15	58.44	21.83	Sharp Price Increase	2020-09-14	199,089,178	2,091,783,870	950.68	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2020	2020-12-15	53.20	2020-12-16	57.57	8.21	Sharp Price Increase	2020-12-15	72,839,808	631,850,374	767.45	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2021	2021-01-12	73.27	2021-01-13	93.33	27.38	Sharp Price Increase	2021-01-12	185,826,731	3,621,641,697	1,848.93	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2021	2021-03-18	72.39	2021-03-19	81.12	12.06	Sharp Price Increase	2021-03-18	191,858,441	2,018,455,187	952.05	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2021	2021-06-22	83.54	2021-06-23	97.46	16.66	Sharp Price Increase	2021-06-22	240,648,274	5,890,471,292	2,347.75	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2021	2021-07-30	89.11	2021-08-02	98.39	10.41	Sharp Price Increase	2021-07-30	205,395,584	1,723,064,957	738.90	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2022	2022-01-20	111.00	2022-01-21	121.00	9.01	Sharp Price Increase	2022-01-20	144,343,100	1,239,546,000	758.75	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2023	2023-01-13	157.00	2023-01-16	187.00	19.11	Sharp Price Increase	2023-01-13	126,643,000	1,263,285,900	897.52	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2023	2023-08-18	177.00	2023-08-21	202.00	14.12	Sharp Price Increase	2023-08-18	202,882,300	2,253,130,800	1,010.56	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase
2025	2025-09-18	555.00	2025-09-19	650.00	17.12	Sharp Price Increase	2025-09-18	602,178,800	4,942,216,800	720.72	Abnormally Large Volume Increase	Sharp Pr Increase Large V Increase

Note. A simultaneous anomaly occurs when the price-anomaly and trading-volume-anomaly criteria are both satisfied on the same retained trading observation.

Table S8. Full event- and news-based contextual-validation audit

Event No.	Event Date	Daily Stock Return (%)	Daily Trading Volume Change (%)	Observed Joint Anomaly	Identified Event or News Context	Contextual Correspondence	Evidence Strength	Sources
2	20 Apr 2016	−16.66	−99.00	Sharp Price Decrease + Large Volume Decrease	No sufficiently specific and temporally eligible issuer, regulatory, project, commodity, or market information was identified within the prespecified search protocol.	External documentary evidence was insufficient to establish a credible contemporaneous correspondence.	Not Externally Validated	—
4	19 Oct 2016	+26.00	+795,118.79	Sharp Price Increase + Large Volume Increase	Contemporaneous Bakrie-group trading strength occurred against a background of elevated coal prices; t+1 reporting documented a 35% rise in BUMI accompanied by the 26% BRMS increase on the preceding day.	The evidence is directionally consistent with the positive BRMS anomaly but is indirect and group/commodity based rather than BRMS-specific.	Moderate	[C1]
5	9 Jan 2017	+21.44	≈−100.00	Sharp Price Increase + Large Volume Decrease	Same-day reporting documented strong BRMS–BUMI price co-movement, but no sufficiently specific contemporaneous public-information catalyst was established.	Co-movement alone does not adequately explain the unusual opposite-direction price–volume configuration.	Not Externally Validated	—
7	26 Jan 2017	+20.80	≈−100.00	Sharp Price Increase + Large Volume Decrease	BUMI-related LQ45, restructuring, and coal-price information coincided with a positive market session in which BRMS was reported among the most actively traded securities.	Group-level information is consistent with the positive price movement, but the external trading-activity description is not readily consistent with the fixed large-volume-decrease classification.	Moderate	[C2], [C3]
11	11 Jan 2018	+22.21	+4,062.96	Sharp Price Increase + Large Volume Increase	BRMS rose sharply alongside other Bakrie-group shares during positive sentiment surrounding ENRG's debt restructuring; t+1 reporting documented both the BRMS price surge and unusually high trading activity.	Price and volume directions correspond closely with the documented group-level sentiment, although the underlying restructuring was not BRMS-specific.	Moderate	[C4]
12	30 Oct 2019	+11.55	+747.72	Sharp Price Increase + Large Volume Increase	BRMS reported positive 9M 2019 results, reversing the prior-year loss, alongside expectations for the commencement of trial gold production at Poboya.	Direct issuer-specific financial and operational information was publicly documented on the anomaly date and was directionally consistent with the positive price–volume event.	High	[C5], [C6]
14	15 Sep 2020	+21.83	+950.68	Sharp Price Increase + Large Volume Increase	No qualifying contemporaneous issuer-specific information was established. Positive BRMS first-half results had been reported on 7 September, outside the prespecified window.	The nearby corporate information is economically relevant but temporally ineligible under the fixed protocol.	Not Externally Validated	—
16	13 Jan 2021	+27.38	+1,848.93	Sharp Price Increase + Large Volume Increase	No sufficiently specific new BRMS disclosure was identified for 10–13 January. Rights-issue information was already public before the window and the related corporate-action process occurred later.	Available documentary evidence does not establish a temporally eligible external correspondence.	Not Externally Validated	—
20	21 Jan 2022	+9.01	+758.75	Sharp Price Increase + Large Volume Increase	The official BRMS announcement that the second rights issue had been completed was dated 17 January, one calendar day before the prespecified window; same-day monetary-policy information was broad rather than issuer-specific.	The most plausible company-specific information fails the predetermined timing criterion.	Not Externally Validated	—

Event No.	Event Date	Daily Stock Return (%)	Daily Trading Volume Change (%)	Observed Joint Anomaly	Identified Event or News Context	Contextual Correspondence	Evidence Strength	Sources
21	16 Jan 2023	+19.11	+897.52	Sharp Price Increase + Large Volume Increase	Indonesian equities strengthened following the release of a trade surplus and favorable global inflation expectations; BRMS was reported among the day's leading gainers at 19.11%.	The broad market environment is directionally consistent with the positive anomaly, but no BRMS-specific catalyst or unique explanation for the extreme volume increase was established.	Moderate	[C8], [C9]
23	19 Sep 2025	+17.12	+720.72	Sharp Price Increase + Large Volume Increase	Ahead of the 19 September index transition, VanEck's 17 September pro forma analysis explicitly identified BRMS as a new MarketVector Global Gold Miners Index constituent at 0.61%; contemporaneous market research anticipated index-related passive inflows.	The issuer-specific index inclusion was publicly known before the event and corresponds closely with the sharp price increase and exceptionally heavy trading on the implementation date.	High	[C10], [C13], [C12]

Note. The primary contextual-search window spans t−3 to t calendar days. Publications at t+1 to t+2 are used only when documenting information already public on or before t. Contextual correspondence is descriptive and does not establish causal attribution.

Supplementary Web-Source Audit List

Source codes [C] and [A] are retained only as an audit key for the supplementary contextual-validation table; conventional author–year citations are used in the main manuscript.

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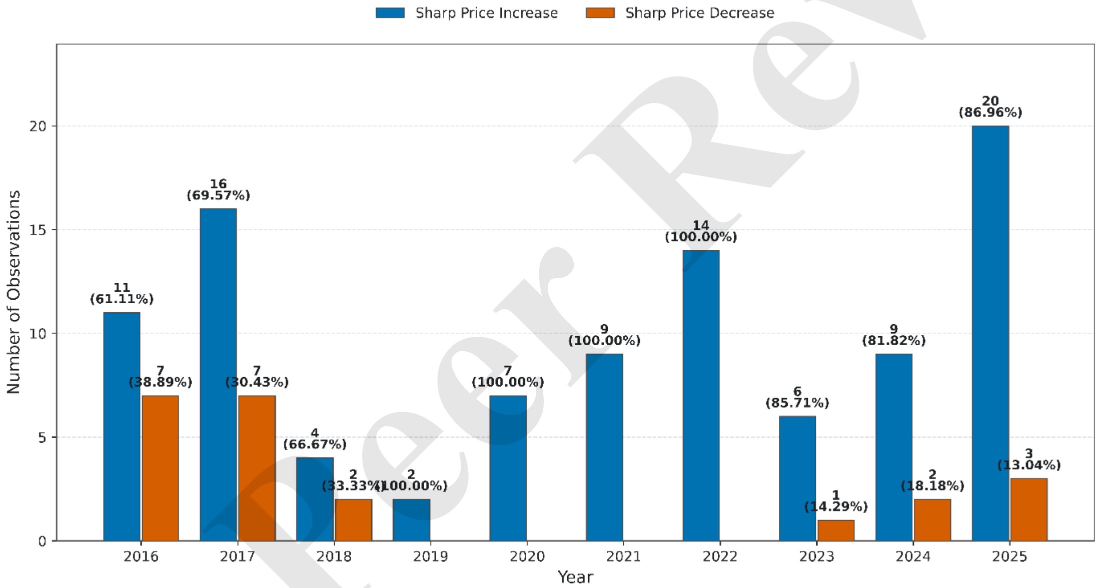
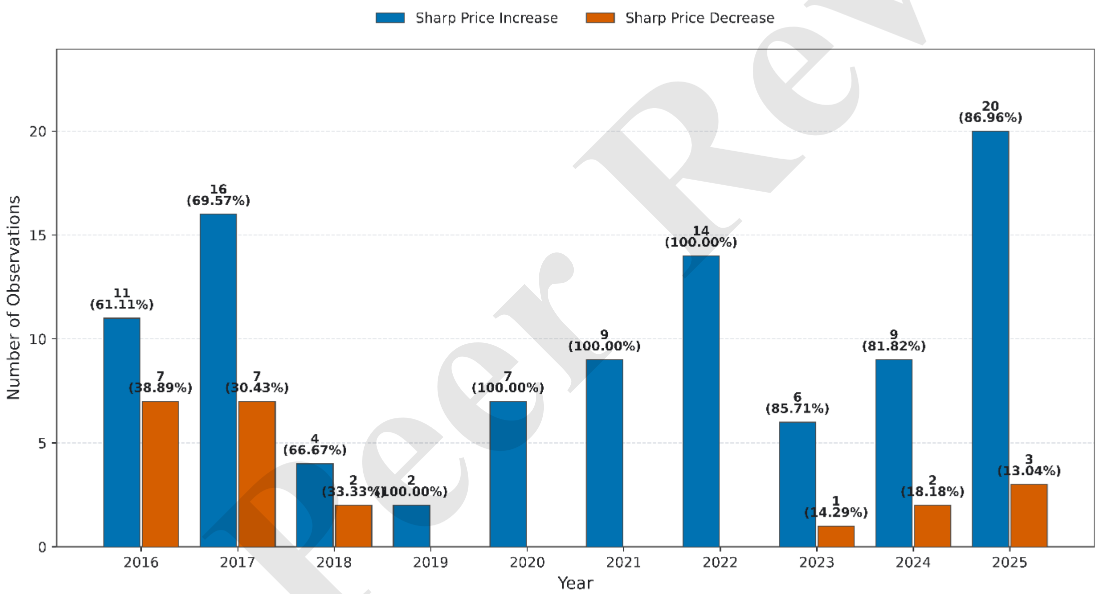
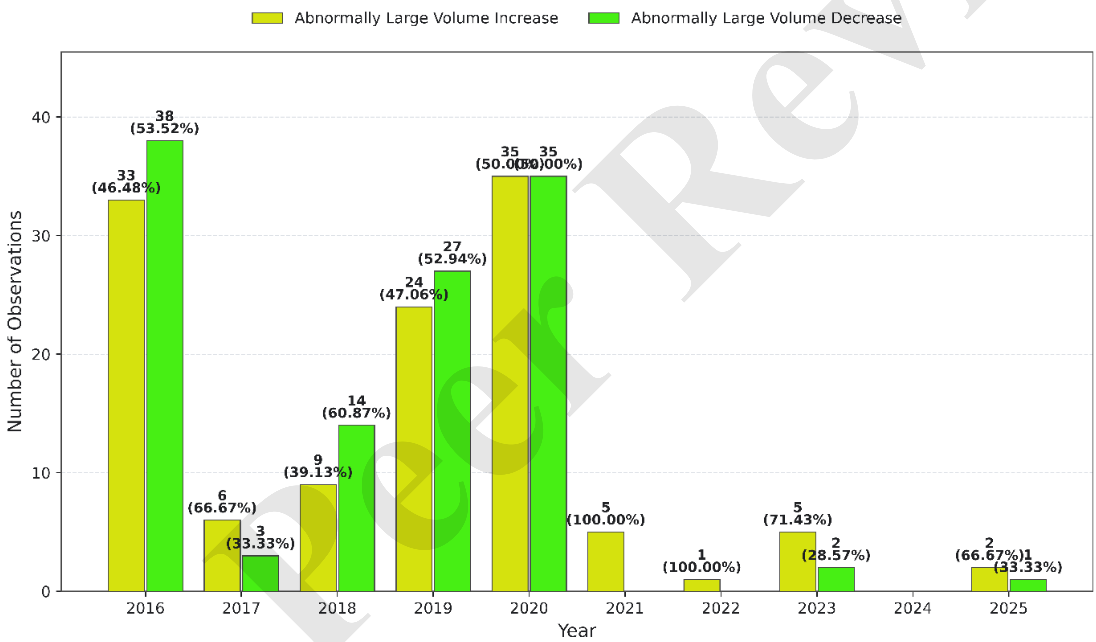
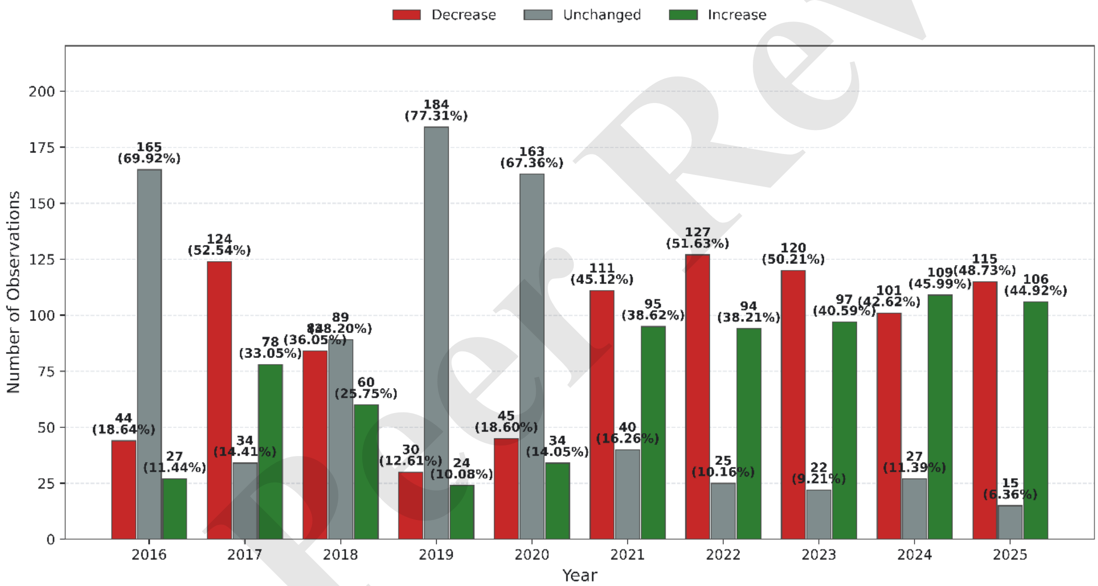

Sharp Price Increase
Sharp Price Decrease
Number of Observations
Year
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
0
5
10
15
20
11 (61.11%)
7 (38.89%)
16 (69.57%)
7 (30.43%)
4 (66.67%)
2 (33.33%)
2 (100.00%)
7 (100.00%)
9 (100.00%)
14 (100.00%)
6 (85.71%)
1 (14.29%)
9 (81.82%)
2 (18.18%)
20 (86.96%)
3 (13.04%)

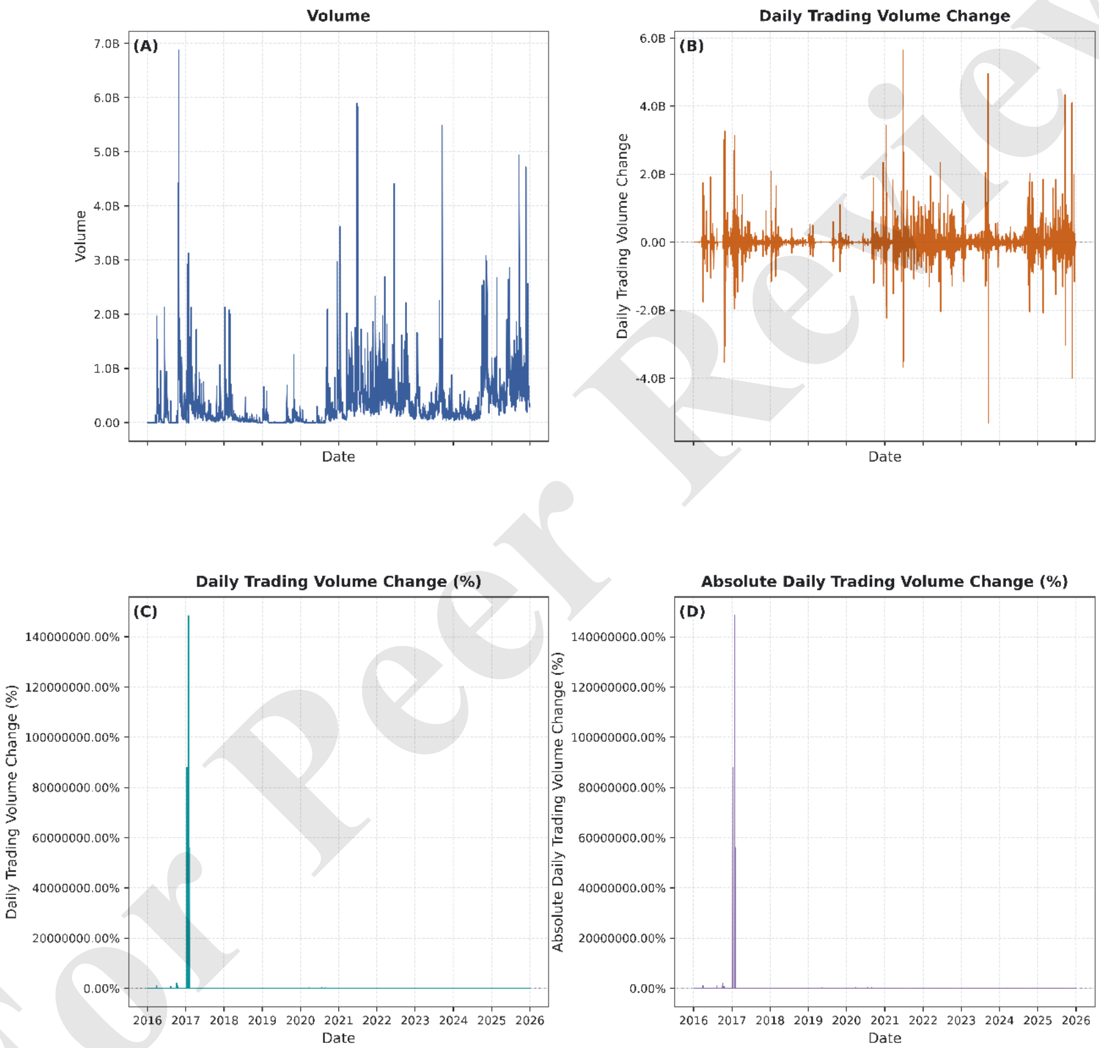

Sharp Price Increase
Sharp Price Decrease
Number of Observations
Year
0
5
10
15
20
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
11 (61.11%)
7 (38.89%)
16 (69.57%)
7 (30.43%)
4 (66.67%)
2 (33.33%)
2 (100.00%)
7 (100.00%)
9 (100.00%)
14 (100.00%)
6 (85.71%)
1 (14.29%)
9 (81.82%)
2 (18.18%)
20 (86.96%)
3 (13.04%)

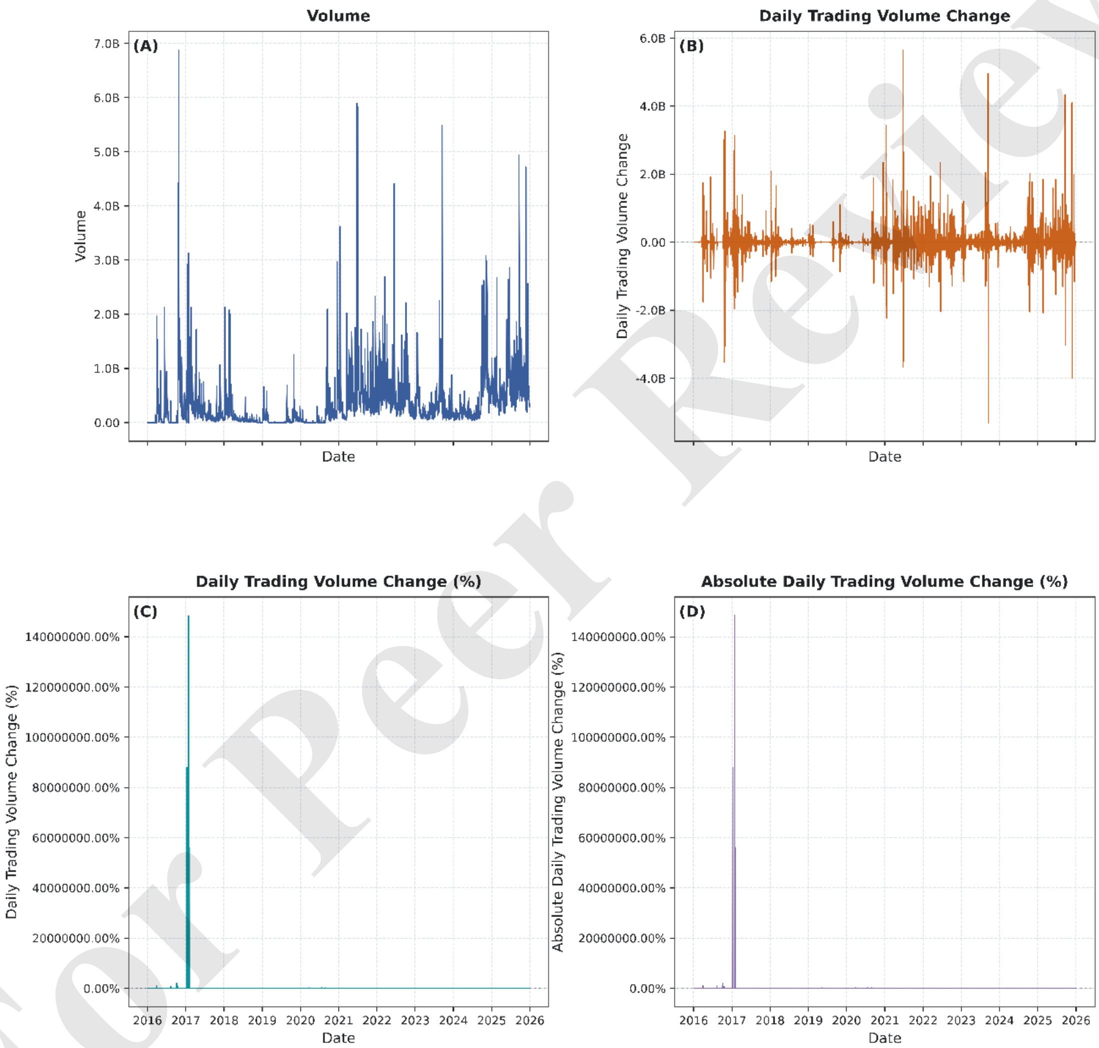


Abnormally Large Volume Increase
Abnormally Large Volume Decrease
Number of Observations
Year

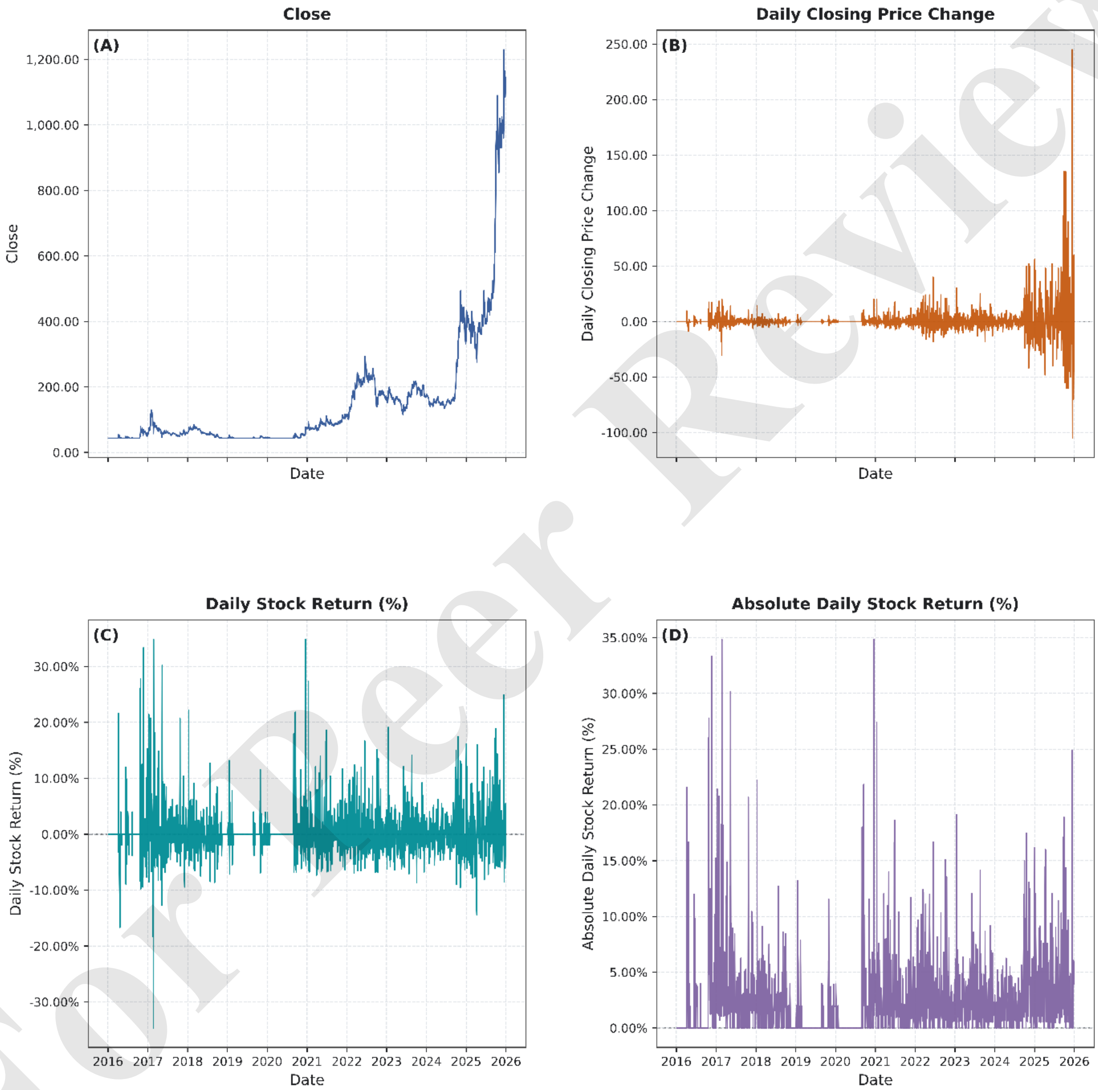


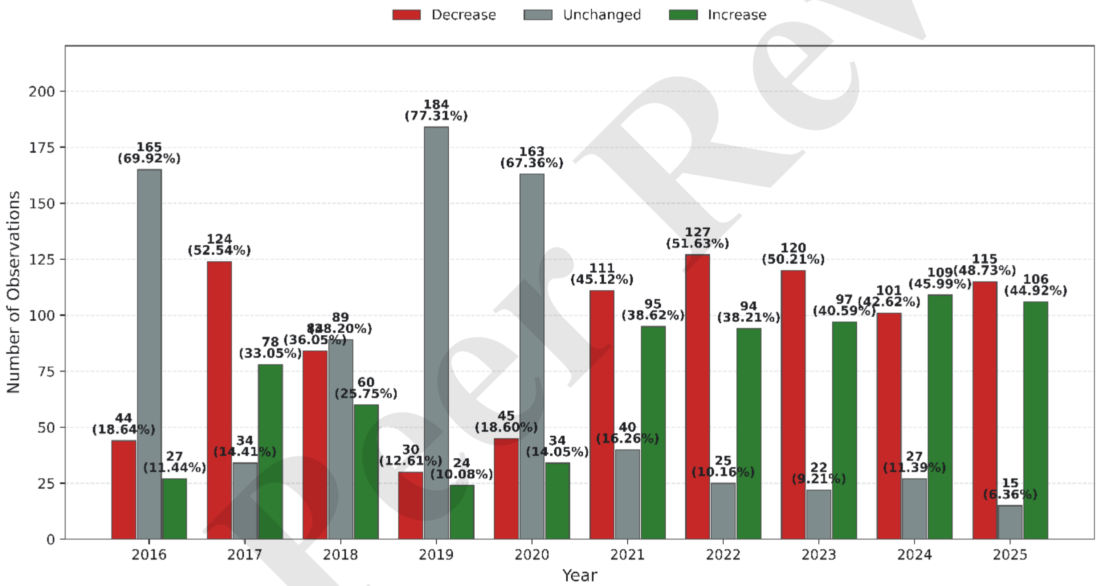
Decrease
Unchanged
Increase
Number of Observations
Year
0
25
50
75
100
125
150
175
200
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
44 (18.64%)
165 (69.92%)
27 (11.44%)
124 (52.54%)
34 (14.41%)
78 (33.05%)
(36.05%)
89 (38.20%)
60 (25.75%)
30 (12.61%)
184 (77.31%)
24 (10.08%)
45 (18.60%)
163 (67.36%)
34 (14.05%)
111 (45.12%)
40 (16.26%)
95 (38.62%)
127 (51.63%)
25 (10.16%)
94 (38.21%)
120 (50.21%)
22 (9.21%)
97 (40.59%)
101 (42.62%)
27 (11.39%)
109 (45.99%)
115 (48.73%)
15 (6.36%)
106 (44.92%)

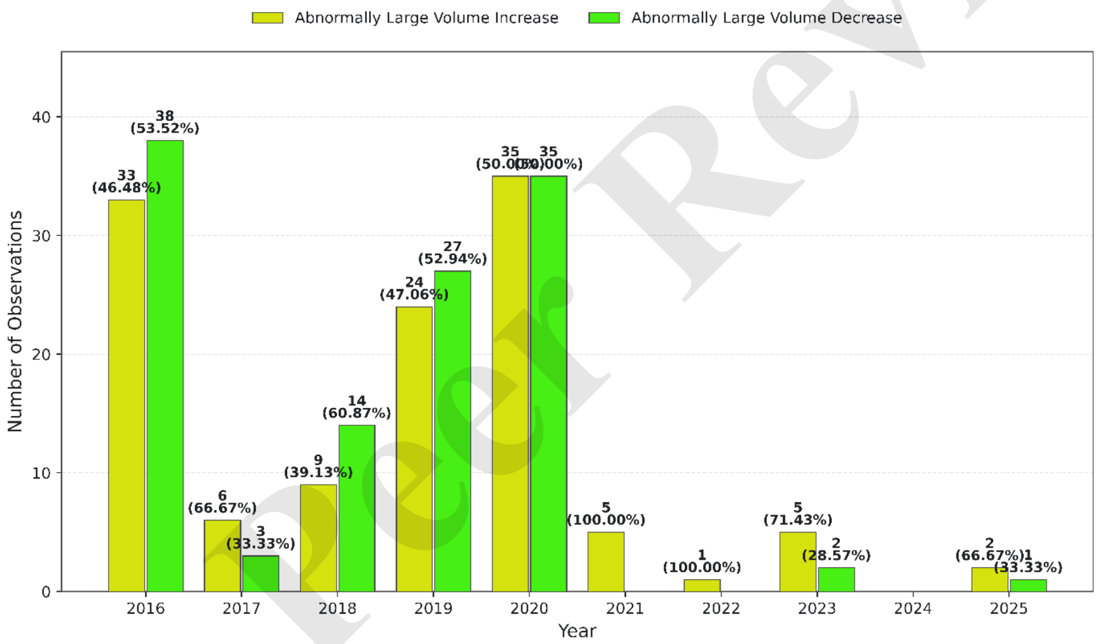

Volume
(A)
7.0B
6.0B
5.0B
4.0B
3.0B
2.0B
1.0B
0.00
Volume
Date
Daily Trading Volume Change
(B)
6.0B
4.0B
2.0B
0.00
-2.0B
-4.0B
Daily Trading Volume Change
Date
Daily Trading Volume Change (%)
(C)
140000000.00%
120000000.00%
100000000.00%
80000000.00%
60000000.00%
40000000.00%
20000000.00%
0.00%
Daily Trading Volume Change (%)
2016 2017 2018 2019 2020 2021 2022 2023 2024 2025 2026
Date
Absolute Daily Trading Volume Change (%)
(D)
140000000.00%
120000000.00%
100000000.00%
80000000.00%
60000000.00%
40000000.00%
20000000.00%
0.00%
Absolute Daily Trading Volume Change (%)
2016 2017 2018 2019 2020 2021 2022 2023 2024 2025 2026
Date



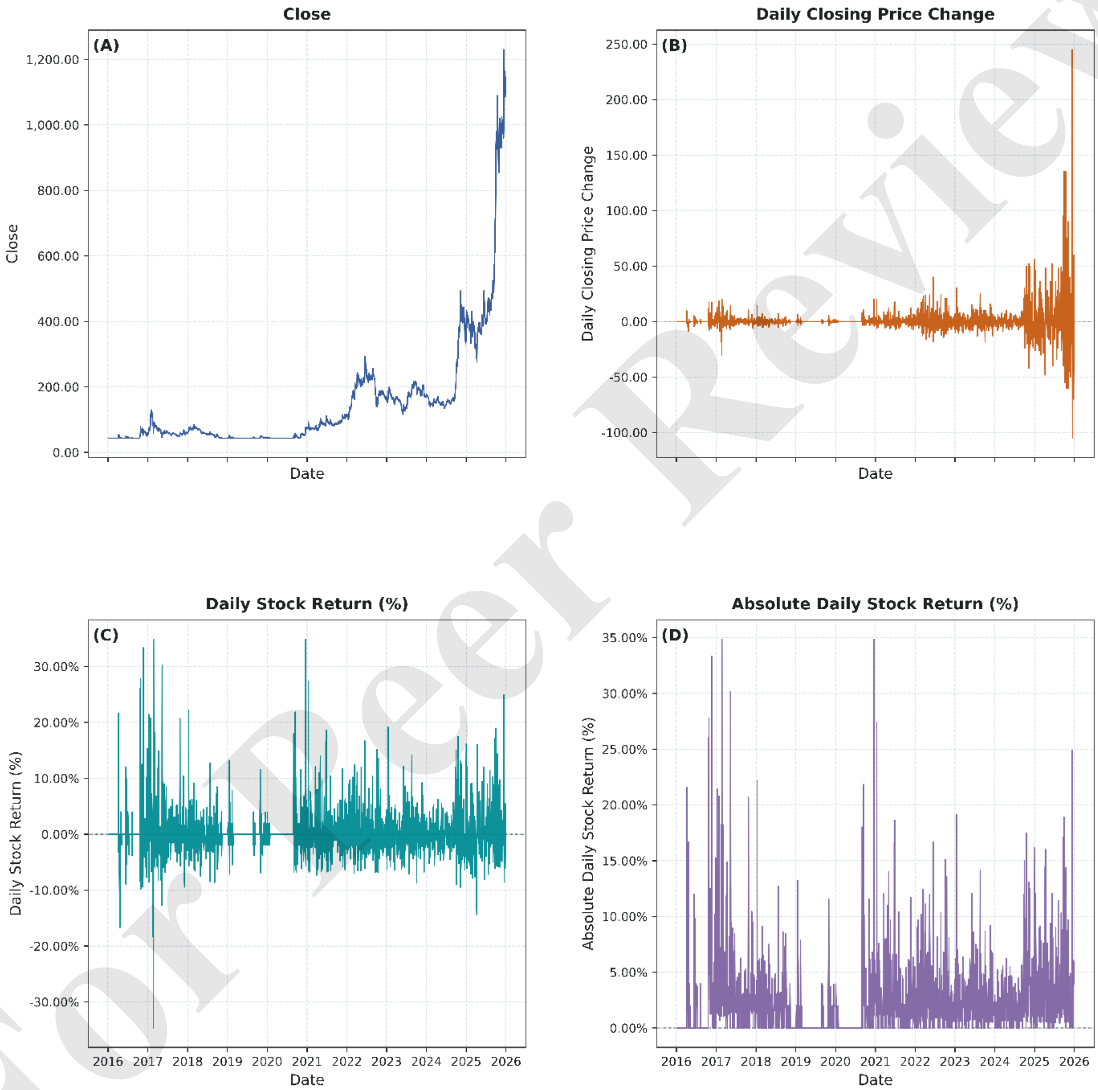
Volume
(A)
7.0B
6.0B
5.0B
4.0B
3.0B
2.0B
1.0B
0.00
Volume
Date
Daily Trading Volume Change
(B)
6.0B
4.0B
2.0B
0.00
-2.0B
-4.0B
Daily Trading Volume Change
Date
Daily Trading Volume Change (%)
(C)
140000000.00%
120000000.00%
100000000.00%
80000000.00%
60000000.00%
40000000.00%
20000000.00%
0.00%
Daily Trading Volume Change (%)
2016 2017 2018 2019 2020 2021 2022 2023 2024 2025 2026
Date
Absolute Daily Trading Volume Change (%)
(D)
140000000.00%
120000000.00%
100000000.00%
80000000.00%
60000000.00%
40000000.00%
20000000.00%
0.00%
Absolute Daily Trading Volume Change (%)
2016 2017 2018 2019 2020 2021 2022 2023 2024 2025 2026
Date


Close
(A)
1,200.00
1,000.00
800.00
600.00
400.00
200.00
0.00
Close
Date
Daily Closing Price Change
(B)
250.00
200.00
150.00
100.00
50.00
0.00
-50.00
-100.00
Daily Closing Price Change
Date
Daily Stock Return (%)
(C)
30.00%
20.00%
10.00%
0.00%
-10.00%
-20.00%
-30.00%
Daily Stock Return (%)
2016 2017 2018 2019 2020 2021 2022 2023 2024 2025 2026
Date
Absolute Daily Stock Return (%)
(D)
35.00%
30.00%
25.00%
20.00%
15.00%
10.00%
5.00%
0.00%
Absolute Daily Stock Return (%)
2016 2017 2018 2019 2020 2021 2022 2023 2024 2025 2026
Date


Decrease
Unchanged
Increase
Number of Observations
Year
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025



Abnormally Large Volume Increase
Abnormally Large Volume Decrease
Number of Observations
Year


Close
(A)
1,200.00
1,000.00
800.00
600.00
400.00
200.00
0.00
Close
Date
Daily Closing Price Change
(B)
250.00
200.00
150.00
100.00
50.00
0.00
-50.00
-100.00
Daily Closing Price Change
Date
Daily Stock Return (%)
(C)
30.00%
20.00%
10.00%
0.00%
-10.00%
-20.00%
-30.00%
Daily Stock Return (%)
2016 2017 2018 2019 2020 2021 2022 2023 2024 2025 2026
Date
Absolute Daily Stock Return (%)
(D)
35.00%
30.00%
25.00%
20.00%
15.00%
10.00%
5.00%
0.00%
Absolute Daily Stock Return (%)
2016 2017 2018 2019 2020 2021 2022 2023 2024 2025 2026
Date